Modelling and Simulation of Mechatronic Systems

Complete Study Notes

Based on A. Macchelli's lectures — UniBO DISI

Study Notes

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Part I

Foundations of Bond Graph Modelling

Chapter 1

Physical Modelling and the Bond Graph Philosophy

The goal of this course is to build **mathematical models of dynamic mechatronic systems** — systems that combine mechanics, electronics, hydraulics and thermodynamics into a single interacting whole[cite: 1]. The challenge is that each of these domains has historically developed its own graphical notation and set of equations (circuit diagrams, free-body diagrams, hydraulic schematics), which makes multi-domain systems hard to model coherently[cite: 2]. **Bond graph theory** solves this by providing a single, unified, energy-based language that describes all physical domains on the same footing[cite: 3]. This uniform notation allows one to study the structure of the system, meaning the nature of the parts and the sets of interconnections among them[cite: 120, 121].

1.1 The systems viewpoint

A **system** is any entity that can be conceptually separated from its environment by a physical or conceptual boundary[cite: 4, 72]. Inside the boundary are interacting subsystems and components; across the boundary flow the stimuli (inputs) and responses (outputs)[cite: 5]. The key insight of the systems viewpoint is that a system's behaviour *as a whole* is not simply the superposition of its parts in isolation — interfaces and interactions matter just as much as individual component dynamics[cite: 6, 84]. In engineering practice, design tasks are broken into parts worked on in isolation, but those parts must ultimately interact[cite: 7, 81]. If the assumptions made in one group are too crude, the assembled system fails to meet its specification[cite: 8]. The systems engineer must hold both the component detail and the system-level behaviour in view simultaneously[cite: 9]. Achieving this ideal goal requires knowledge pertinent to the study of systems per se, bridging isolated disciplines like vibrations, dynamics, fluid mechanics, thermo-dynamics, and automatic control[cite: 92, 93, 94, 95, 96, 97].

Key Insight: The modelling trade-off

No system can be modelled exactly. The art is to choose the **right level of complexity**: detailed enough to answer the questions being asked, simple enough to be tractable[cite: 10, 74]. A model that is too simple will miss important dynamics[cite: 11]; one that is too detailed will obscure the physics and be computationally intractable[cite: 12]. This judgement is acquired through practice and physical intuition[cite: 13].

Dynamic models serve four distinct purposes:

- **Analysis**: Predict the future given known inputs and initial conditions[cite: 14]. This can be data-based (black box) or physics-law based (white box) depending on how the component is characterized[cite: 201, 212, 214].
- **Identification**: Find the model from measured input/output histories[cite: 14].
- **Synthesis**: Find the system that turns a given input into a desired output[cite: 14].
- **Control**: Design a feedback law that drives the system to a desired behaviour[cite: 14].

Each task uses the same model but in a different way[cite: 14].

1.2 Power as the universal currency

The central idea of bond graph theory is that **all physical interactions involve the exchange of power**[cite: 15, 229]. When a motor drives a pump, when a spring pushes a mass, when a voltage source drives a

current through a resistor — in every case, energy flows across a boundary, and the instantaneous rate of that flow is power[cite: 16]. By making power the primary language, bond graphs work equally well in every energy domain[cite: 17]. At every interconnection point between subsystems — called a **port** — two variables are defined whose product is instantaneous power:

$$P(t) = e(t)\, f(t) \tag{1.1}$$

These are called the **effort** e and the **flow** f[cite: 18]. Their specific physical meaning depends on the domain[cite: 19]:

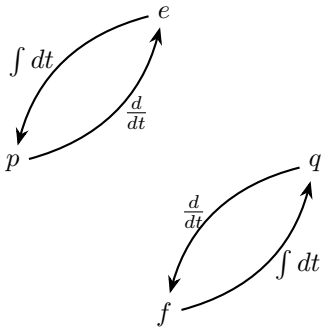
Domain	**Effort** e	**Flow** f	**Power** ef
Mechanical (translation)	Force F [N]	Velocity V [m/s]	[W]
Mechanical (rotation)	Torque τ [N·m]	Ang. velocity ω [rad/s]	[W]
Electrical	Voltage e [V]	Current i [A]	[W]
Hydraulic	Pressure P [Pa]	Volume flow Q [m^3/s]	[W]

Two additional variables complete the picture[cite: 21]. The **momentum** p (generalised) is the time integral of effort[cite: 21]; the **displacement** q is the time integral of flow[cite: 22]:

$$p(t) = p_0 + \int_{t_0}^{t} e(\tau)\, d\tau, \qquad q(t) = q_0 + \int_{t_0}^{t} f(\tau)\, d\tau, \qquad \dot{p} = e, \quad \dot{q} = f. \tag{1.2}$$

These are the **energy variables** (or state variables)[cite: 23, 330]. While power is the instantaneous flow, energy flow is the time integral of that power: $E(t) = \int^t P(\tau)d\tau = \int^t e(\tau)f(\tau)d\tau$[cite: 340, 342]. The reason for their name is that energy stored in a system is always a function of these integrated quantities — for a spring, $E = kq^2/2$; for a mass, $E = p^2/(2m)$; for a capacitor, $E = q^2/(2C)$; for an inductor, $E = p^2/(2L)$[cite: 23, 24]. In cases where an effort is a function of displacement, the energy is $E(q) = \int e(q)dq$; conversely, when flow is a function of momentum, $E(p) = \int f(p)dp$[cite: 344, 345, 348].

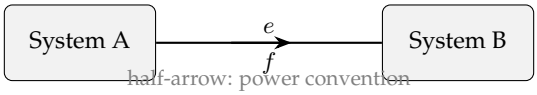
The relationship between these four variables is elegantly summarized by the **Tetrahedron of State**:



The power and energy variables — $\{e, f, p, q\}$ — are the only types needed to model physical systems in the bond graph framework[cite: 25, 347].

1.3 Bonds, ports and multiports

A **subsystem with one or more ports** is called a **multiport**[cite: 26]. Each port is a place where power can flow in or out[cite: 27]. Familiar examples: an electric motor is a 3-port (field winding, armature winding, mechanical shaft)[cite: 28]; a hydraulic pump is a 2-port (shaft, fluid line); a mechanical spring is a 1-port[cite: 29]. When two multiports are connected — meaning one port of each is joined — power flows between them and the effort and flow at the shared port become identical for both[cite: 30, 488]. This shared port is a **bond**, drawn as a line with a half-arrow indicating the assumed positive direction of power flow[cite: 31]:



When major sub-systems are represented by words instead of internal details, the resulting graph is known as a **word bond graph**[cite: 509, 514]. The half-arrow is *not* the direction of causality and *not necessarily* the direction energy actually flows at any given instant — it simply sets the sign convention: power is positive when it flows in the direction of the arrow[cite: 34].

Definition: Bond

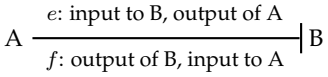
A bond is a line connecting two ports[cite: 35]. It carries *both* an effort e and a flow f simultaneously[cite: 35]. The product ef is the instantaneous power flowing in the direction of the half-arrow[cite: 36]. A bond is a 2-wire connection in the metaphorical sense: it always transmits a pair of conjugate variables[cite: 37].

1.4 Causality: inputs, outputs and causal strokes

At any port, one of the two power variables is the **input** (cause) and the other is the **output** (effect)[cite: 38]. One can either impose a force and observe the resulting velocity, or impose a velocity and observe the resulting force — but **never both simultaneously**[cite: 39, 537]. This choice is called the **causality** of the port[cite: 40].

Causality is indicated on a bond graph by a **causal stroke**: a short perpendicular bar at one end of the bond[cite: 41, 581]. It is critical to remember that the half-arrow power direction indications and the causal stroke are **totally uncorrelated**. The rule is:

- The causal stroke marks the end toward which the **effort** is being directed (effort is an input to that end)[cite: 41, 581].
- The **flow** signal is then directed *away* from the stroked end (flow is an output of that end)[cite: 42, 582].



Causal assignment is not free — each type of element has constraints on which causalities are admissible[cite: 46]. The choice of causality throughout the bond graph propagates and constrains subsequent choices, and ultimately determines whether the system equations are explicit ODEs, implicit DAEs, or contain algebraic loops[cite: 47]. This is detailed in Chapter 4[cite: 48].

1.5 Pure signal flow and active bonds

Not all interconnections transmit significant power[cite: 49]. A sensor measuring a velocity transmits negligible power back to the mechanism it measures[cite: 49]; an electronic amplifier takes a tiny input signal and produces a much larger output, with power supplied by an external source[cite: 50]. In these cases, one of the two power variables is negligible, and the bond **degenerates to a single signal** — called an **active bond**, drawn with a full arrow rather than a half-arrow[cite: 51, 613]. Because their description is functional rather than physical, ideal instruments and amplifiers represented by these bonds violate many physical laws[cite: 635, 637]. Active bonds are used to represent signal processing, controllers, and ideal sensors within an otherwise power-based bond graph[cite: 52]. The combination of bond graphs (for the physical plant) and block diagrams (for the controller) is the natural language for mechatronic system design[cite: 53, 670].

Chapter 2

Basic Bond Graph Elements

A remarkably small set of ideal elements suffices to model the dynamics of systems across all energy domains. There are **nine** fundamental bond graph elements: three 1-port passive elements (R, C, I), two source elements (Se, Sf), two 2-port elements (TF, GY), and two junction elements (0-junction, 1-junction). Every physical system model is assembled from these building blocks.

Real devices can be considered as subsystems from the point of view of power exchanges and external port variables. These basic multiport elements are idealized mathematical models—or conceptual "entities"—used to model specific physical effects within a device. The bond graph notation often allows one to visualize aspects of the system structure more easily than with state equations or graphical notations designed strictly for a single energy domain.

2.1 The 1-port resistor *R*: energy dissipation

A resistor is a 1-port in which effort and flow are related by a **static constitutive law** $e = \mathcal{R}(f)$ (or equivalently $f = \mathcal{R}^{-1}(e)$). No energy is stored: whatever power enters the port is dissipated instantly (converted to heat in the thermal domain, which is usually neglected in the model). In the e–f plane, a passive resistor's constitutive curve lies in the first and third quadrants, ensuring $ef \geq 0$ (power always flows in). Power flows into the port when the product of e and f is positive according to the assigned sign convention. Because energy seems to disappear from the electrical, mechanical, or hydraulic domains, it is in reality transferred into the thermal domain, whose presence is typically neglected in these basic dynamic models.

Physical realisations: electrical resistance $e = Ri$; viscous damper $F = bV$; fluid resistance $\Delta P = R_f Q$ (linear); orifice flow $\Delta P = kQ|Q|$ (nonlinear).

In the bond graph:



The half-arrow points toward R (positive power convention). For a linear R: appending `R:R` (or `R:b` for a damper) names the parameter.

It is important to note that R elements are not only used as pure dissipation elements. Sometimes they act as active components where an algebraic relationship between effort and flow is strictly required to characterize the device's behavior.

2.2 The 1-port capacitor *C*: potential energy storage

A capacitor (or compliance) stores **potential energy** via a static relationship between effort and displacement:

$$e = \mathcal{C}^{-1}(q), \qquad q = \int f\, dt. \tag{2.1}$$

In the linear case, $e = q/C$. The linear compliance C is defined as the inverse of the slope of the e versus q constitutive curve.

Stored energy:

$$E(q) = \int_{q_0}^{q} e(\bar{q})\, d\bar{q} \quad \longrightarrow \quad E = \frac{q^2}{2C} \text{ (linear)}. \tag{2.2}$$

Since the capacitor stores energy without loss, the power flow into the port can reverse, meaning power flows back out into the system. During this process, energy is strictly conserved.

Physical realisations: electrical capacitor $e = q/C$; spring $F = kx$ (so $C = 1/k$); hydraulic accumulator $P = V/C_a$; torsional spring $\tau = k_r\theta$.

Key Insight: Preferred causality of C

The natural (preferred) causality for a capacitor is **flow-in, effort-out**: flow is integrated to give displacement, from which effort is computed. This is called *integral causality*. If the bond graph forces effort-in on C (derivative causality), the element does not contribute an independent state variable, which has implications for the system's order.
Intuitively, from a simulation perspective, you impose the flow (e.g., current or velocity) which the element integrates into displacement (charge or position), and this displacement yields the effort (voltage or force) as an output. Conversely, imposing effort requires differentiating the displacement to get flow, which leads to derivative causality.

2.3 The 1-port inertia *I*: kinetic energy storage

An inertia (inductor) stores **kinetic energy** via a static relationship between flow and momentum:

$$f = \mathcal{I}^{-1}(p), \qquad p = \int e\,dt. \tag{2.3}$$

In the linear case, $f = p/I$. Here, the linear inertia I is defined as the inverse of the slope of the f versus p curve.

Stored energy:

$$E(p) = \int_{p_0}^{p} f(\bar{p})\,d\bar{p} \quad \longrightarrow \quad E = \frac{p^2}{2I} \text{ (linear)}. \tag{2.4}$$

Physical realisations: point mass $F = ma \Rightarrow V = P/m$ (so $I = m$); rotating body $\tau = J\dot{\omega} \Rightarrow \omega = p_\tau/J$; fluid inertance $\Delta P = I_f\dot{Q}$.

Key Insight: Preferred causality of I

The natural causality for an inertia is **effort-in, flow-out**: effort is integrated to give momentum, from which flow is computed. Again, this is integral causality. Newton's second law is a direct statement of this: force (effort) integrates to momentum, from which velocity (flow) is recovered.
As with the capacitor, imposing flow on an inertia leads to derivative causality because the imposed flow must be differentiated to find the required effort, thus reducing the number of independent state variables.

The tetrahedron of state variables (Fig. on slide 15, pack M01) summarises the four types and their relationships: $\dot{p} = e$, $\dot{q} = f$, with C linking e–q and I linking f–p.

2.4 Source elements *Se* and *Sf*

An **effort source** Se maintains a specified effort $e(t)$ at its port, regardless of the flow (infinite power if needed — the idealisation of a stiff voltage/force/pressure supply). A **flow source** Sf dually maintains a specified flow. Their causality is fixed: Se always imposes effort (causal stroke away from Se); Sf always imposes flow.

In reality, components always find a compromise. For instance, when you set the voltage of a real battery, the act of drawing current inherently lowers the voltage itself. Because ideal sources do not model this compromise in one go, a real-world component like a battery must be modeled using an ideal voltage source (Se) in series with a resistance (R) to accurately simulate the voltage drop under load.

When the source value is itself a function of a signal from another part of the graph, it becomes a **modulated source** MSe or MSf. This is how controlled actuators, variable-flow pumps, and vibration shakers are represented.

2.5 The transformer *TF*: effort–effort, flow–flow exchange

A transformer is a **power-conserving 2-port**: $e_1 f_1 = e_2 f_2$ at all times. Its constitutive relations scale effort and flow by a constant (or time-varying) modulus m:

$$e_1 = m\,e_2, \qquad m\,f_1 = f_2. \tag{2.5}$$

Effort is scaled up by m, flow is scaled down by m (or vice versa). Physical examples: electrical transformer ($m = N_1/N_2$, turns ratio); lever ($m = b/a$, arm ratio); hydraulic piston ($F = AP$, $AV = Q$, so $m = A$); gear pair ($m = r_1/r_2$); rack and pinion.

Key Insight: TF causality

Two causal forms are valid for TF: either effort-in on port 1 (and effort-out on port 2), or effort-in on port 2. The causal stroke must appear on exactly one of the two ports. If the geometry changes (e.g. a linkage), m becomes time-dependent — the element becomes a **modulated transformer** MTF.
In robotics, the Jacobian matrix plays the role of the MTF modulus. The Jacobian expresses the relationship between end-effector and joint velocities ($v = J(\theta)\dot{\theta}$). Assuming power conservation, it naturally also relates the end-effector applied force to the joint torques ($\tau = J^T(\theta)F$).

2.6 The gyrator *GY*: effort–flow exchange

A gyrator is also power-conserving, but it exchanges the roles of effort and flow:

$$e_1 = r\, f_2, \qquad r\, f_1 = e_2. \tag{2.6}$$

This is fundamentally different from TF: while a transformer maps effort to effort and flow to flow through a constant, the gyrator maps a flow to an effort and vice versa through a constant. Physical examples: DC motor (back-EMF $e_a = T\omega$, torque $Ti_a = \tau$, so $r = T$); gyroscope (a force in one direction induces an angular velocity in the perpendicular direction).

Key Insight: Gyrator is more fundamental than transformer

The gyrator can sometimes seem like a mysterious element, but it is actually more fundamental than a transformer. Two cascaded gyrators produce a transformer. More profoundly, a gyrator essentially interchanges the roles of effort and flow, so it maps C-elements into I-elements: a capacitor seen through a gyrator behaves as an inertia. This means one does not fundamentally need two types of energy-storage elements — the gyrator generates one from the other. Historically, ignoring gyrators and forcing gyroscopic systems into transformer notation led to incorrect models.

When the gyration ratio r depends on a signal (e.g. motor torque constant varying with field current), the element becomes a modulated gyrator **MGY**.

2.7 Junction elements: 0-junction and 1-junction

Junctions are **power-conserving** multi-port elements that distribute power among connected elements. They enforce algebraic constraints on efforts and flows. Fundamentally, zero and one junctions are simply mathematical **conditions** applied to the system (such as force balance, flow conservation, or parallel branches sharing the same voltage).

2.7.1 The 0-junction: common effort

At a 0-junction all connected bonds share the **same effort**, and the flows sum algebraically to zero (with signs set by the half-arrow directions):

$$e_1 = e_2 = e_3 = \cdots, \qquad \sum_k \pm f_k = 0. \tag{2.7}$$

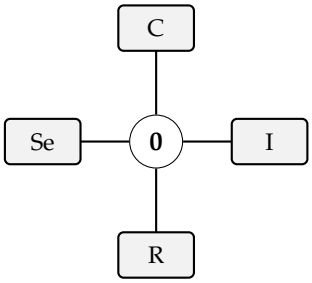
Electrical analogue: a **node** (Kirchhoff's current law, same voltage). Mechanical analogue: a point where a single force acts on several elements with velocities that sum to zero (geometric compatibility — spring and damper in parallel). Here, we are imposing a relation between relative velocities. Hydraulic analogue: a junction of pipes (conservation of volume flow).

2.7.2 The 1-junction: common flow

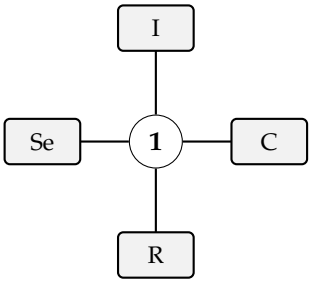
At a 1-junction all connected bonds share the **same flow**, and the efforts sum algebraically to zero:

$$f_1 = f_2 = f_3 = \cdots, \qquad \sum_k \pm e_k = 0. \tag{2.8}$$

Electrical analogue: a **loop** (Kirchhoff's voltage law, same current). Mechanical analogue: Newton's law for a mass — all forces acting on an inertia sharing a single velocity must balance. It encodes the action-reaction principle and effectively acts as a rigid connection among the mechanical elements. Hydraulic analogue: pressure drops around a single-flow circuit sum to zero.



0-junction: same effort



1-junction: same flow

Watch out: Sign conventions at junctions

The algebraic sum at a junction depends on the *direction of the half-arrows*. One convention: at a 0-junction, the half-arrow of the "driving" bond points *toward* the junction, and all others point away — then the sum of all f_k equals the driving flow. Always trace the signs carefully; an error here propagates into wrong equations of motion. Attention: You can change the direction of the bonds if and only if you change the signs in the sum equation accordingly (flows for 0-junctions, efforts for 1-junctions).

2.7.3 Bond graph identities and simplification

A 0-junction with only two bonds connecting to it reduces to direct connection (same effort, same flow). A 1-junction with only two bonds similarly reduces. These **bond graph identities** allow systematic simplification of an initial, verbose bond graph into a compact one. They are used heavily when eliminating zero-power bonds (e.g. after connecting to ground in a mechanical model) and when simplifying "reducible loops" that appear when spring and damper share the same relative velocity.

2.8 Causality summary for basic elements

For effort and flow sources, causality is trivial and fixed. For basic 2-ports (TF, GY), there are two possible causal assignments; as soon as one variable is assigned as an input, the other is constrained by the constitutive relation. Similar logic naturally extends to 3-ports (junctions).

Element	Preferred causality	Alternative	Notes
Se	effort out (fixed)	—	always imposes e
Sf	flow out (fixed)	—	always imposes f
R	either	either	indifferent (usually)
C	flow in (integral)	effort in (derivative)	derivative ⇒ dependent state
I	effort in (integral)	flow in (derivative)	derivative ⇒ dependent state
TF	one of two forms	—	must be one in, one out
GY	both in *or* both out	—	two valid forms
0-jct	one effort in, rest out	—	uniquely determined
1-jct	one flow in, rest out	—	uniquely determined

Chapter 3

System Models in Different Domains

Using the bond graph arsenal, it is possible to create mathematical models for a wide variety of physical systems (but not for all, e.g. thermal systems are typically not included here). With the nine basic elements in hand, we can build bond graph models of electrical circuits, mechanical systems (translational, rotational, planar), and hydraulic circuits. The procedure is systematic and domain-independent; once mastered for one domain, it transfers immediately to all others. We start with single-energy-domain systems, then introduce transducer elements (e.g., motors, sensors), and finally arrive at multi-energy-domain (mechatronic) systems.

3.1 Electrical circuits and networks

Any electrical circuit can be modelled by a bond graph containing elements of the set $\{0, 1, R, C, I, Se, Sf\}$. For electrical circuits the mapping to bond graph variables is direct: voltage is effort, current is flow. The 0-junction encodes Kirchhoff's current law at a node (same voltage, currents sum to zero); the 1-junction encodes Kirchhoff's voltage law around a loop (same current, voltage drops sum to zero).

Construction procedure. Constructing the overall bond graph model is much easier if a procedure can be followed that ensures success regardless of the complexity.

1. **Assign a power convention**: for each circuit element choose positive current direction and positive voltage drop. Talking about power in BG, we typically ensure that, when positive, power is flowing *towards* the passive 1-port element. Note that active elements are elements providing power (sources, but also energy-storing elements freeing their energy). In some situations, sources may even absorb power.
2. **One 0-junction per node voltage**: place a 0-junction for every distinct node voltage (including what will be the ground). Every bond touching a 0-junction carries the same effort (node voltage).
3. **One 1-junction per element**: for each circuit element (R, L, C, source), insert a 1-junction to represent the voltage difference across it. By properly directing the half-arrows on 1-junctions, the voltage drop can be established across each element. Connect the 1-junction between the two 0-junctions at its terminals. Attach the bond graph element (R, I, C, Se, Sf) to the 1-junction.
4. **Ground = zero effort source**: choose a ground node and connect the ground 0-junction to a Sf : 0 (zero-flow source) or Se : 0 (zero-voltage), which in practice means deleting all bonds with zero power at that junction. Note that different bond graphs will result for a given circuit if we choose different ground nodes.
5. **Simplify**: apply bond graph identities to remove redundant junctions and bonds.



An electrical network is an extension of electrical circuits to include idealised transformers and gyrators. Electrical gyrators are exhibited in Hall effect transducers, where voltage across a semiconductor material is related to a current through the material perpendicular to the voltage drop direction. The same construction method extends to **electrical networks** containing transformers (TF) and gyrators (GY). The transformer element inserts between two 1-junctions; the turns ratio N becomes the TF modulus. The basic rules for bond graph construction remain unchanged.

Key Insight: Ladder and T/pi structures

With practice, you will observe certain recurrent patterns, and your ability to do more by inspection will increase. Transmission lines and long hydraulic lines have lumped approximations as ladder networks — cascaded T or pi sections. In bond graph terms, the T-section corresponds to alternating 1-junctions (series inductors) and 0-junctions (shunt capacitors). The topological dual (exchanging 0 and 1 junctions) gives the pi-section. Notice that the difference between these two schemes is the role of current and voltage is interchanged. This technique holds in general when we want to find the topological dual of even complex circuits: exchange 0 and 1 junctions in the BG! The pi and tee structures become truncations of the ladder structure that allow for reasonable inclusion of some of the transmission line dynamics without having the overall system become too large. Understanding this duality is a powerful shortcut when building bond graphs by inspection.

3.2 Mechanical translation

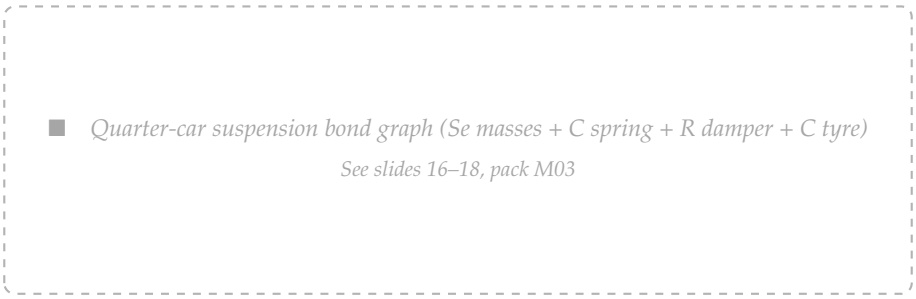
Mechanical systems are those composed of components such as masses, springs, dampers, levers, flywheels, gears, shafts, and so forth. When dynamic systems are put together from these components, we must interconnect rotating and translating inertial elements with axial and rotational springs and dampers, and we must appropriately account for the kinematics of the systems.

For mechanical translation, velocity is flow and force is effort. The analogy with electrical circuits is *dual*: in electrical circuits, nodes share voltage (0-junction); in mechanics, rigid connections share velocity (1-junction).

Construction procedure for Mechanical Translation.

1. **Indicate velocity directions**: on a free-body-type sketch, use arrows and symbols to indicate the positive direction of absolute velocity components. Fixing conventional directions in this scheme is the first crucial step! State whether force-generating elements are positive in tension or compression using symbols such as +T or +C.
2. **One 1-junction per distinct velocity**: each inertia is attached to its absolute velocity 1-junction.
3. **0-junctions for relative velocities**: spring and damper forces react to the rate of extension, meaning they depend on the *relative* velocity across them. A 0-junction connecting two 1-junctions produces the relative velocity difference needed by the compliant/resistive element. The sign at the 0-junction must be chosen so that the relative velocity is positive when the element is in the assumed tension/compression state. When forces are positive in tension, power will flow toward the elements.
4. **Reference (ground)**: the inertial frame is fixed in zero velocity, represented by a Sf : 0, which is deleted after establishing relative velocities.
5. **Attach elements**: inertias on 1-junctions; springs (C : $1/k$) and dampers (R : b) on their relative-velocity 0-junctions; gravity and other absolute forces as Se sources on velocity 1-junctions.

The *beauty* of power conservation in junctions is that we need only to constrain the velocities, and the forces will be automatically balanced without further effort. Newton's second law and the action-reaction principle are built in without explicitly writing them. We can perform a double check to see that the force balance is realized by the 1-junctions.



Reducible loop. When a spring and damper are in parallel (same relative velocity), the 0-junction connecting both their velocity 1-junctions can be simplified using bond graph identities into a single 0-junction carrying both the C and R elements. This is the **reducible loop** — a very common pattern that emphasizes that several elements share exactly the same relative velocity.

Levers, pulleys and kinematic linkages. With minor efforts, it is possible to deal with translational systems containing levers, pulleys, and other simple motion-force transforming devices. These introduce position-dependent force/velocity transformations, represented by modulated transformers (MTF) or fixed TF when the geometry is constant. The MTF modulus is the relevant geometric ratio (e.g. the Jacobian in robotics).

3.3 Fixed-axis rotation

Fixed-axis rotation occurs in the mechanics of rotating machines, such as electric motors and internal combustion engines, gear boxes, transmissions, transfer cases, differentials, and drive shafts.

Only minor modifications to the procedure for mechanical translation allow handling systems with fixed-axis rotation. It is entirely analogous to translation with the variable substitution $V \to \omega$, $F \to \tau$, $m \to J$, $k \to k_r$, $b \to b_r$.

The construction procedure is identical: 1-junctions for distinct absolute angular velocities, 0-junctions for relative angular velocity across compliant or resistive rotational elements, TF for gear pairs (modulus = gear ratio r_1/r_2), and eliminating the bonds with zero angular velocity.

In the fixed-axis rotation framework, it's not really relevant to talk about +T or +C; what is relevant is to maintain coherence with the relative velocities across all the schemes (e.g., choosing one velocity as the upstream master velocity and the other as the downstream slave velocity).



3.4 Plane motion of rigid bodies

Mechanical bodies with finite mass both translate and rotate in real applications; their dynamics are governed by a simultaneous combination of translation and rotation. 3-D motion is not trivial, so plane motion is much simpler. Plane motion results when the inertial bodies of a physical system are constrained to translate in two dimensions and to rotate only about an axis perpendicular to the plane of motion. The motion is always w.r.t. an inertial frame (with eventually coordinate transformations in the middle).

When a body both translates and rotates in a plane, three absolute velocity components are needed: v_x, v_y (translational velocities of the centre of mass) and ω (angular velocity). Newton's law in the plane gives three 1-junctions, one per component, each with its own inertia element:

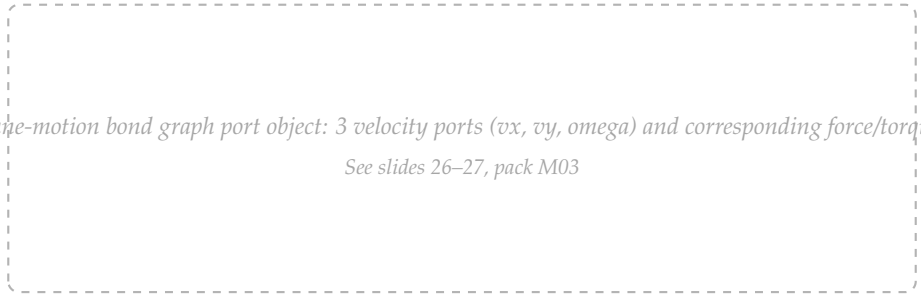
$$m\dot{v}_x = F_x, \quad m\dot{v}_y = F_y, \quad J\dot{\omega} = \tau, \tag{3.1}$$

so the kinetic energy is $T = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}J\omega^2$ with three I-elements. This dynamic representation does not involve directly any position information, which is coherent with the fact that this object stores kinetic energy, not potential energy.

Devices are attached at points P offset from the centre of mass by $(r_P \cos\theta, r_P \sin\theta)$ in the inertial frame. The velocity of point P is:

$$v_{Px} = v_x - (r_P \sin\theta)\omega, \qquad v_{Py} = v_y + (r_P \cos\theta)\omega. \tag{3.2}$$

These are linear combinations of the three body-velocity components — implemented with MTF elements (moduli $r_P \sin\theta$ and $r_P \cos\theta$, time-varying through θ). By power conservation ($\Delta F_x v_x + F_y v_y + \tau\omega = F_{Px}v_{Px} + F_{Py}v_{Py} + \tau_P\omega_P$), the corresponding forces and torques at the body's ports are automatically balanced without writing free-body equations explicitly. The result is an "object" with a force/velocity interface toward the environment (what is outside the system). All the port variables are expressed with respect to the inertial frame.



Key Insight: Parenthesis: the Body Frame!

To characterize the energy of a rigid body, it is necessary to know the velocity of the center of mass and the angular velocity. The velocities discussed so far are expressed in the inertial reference, but for multi-body dynamics, it is sometimes convenient to introduce body-fixed coordinates. This coordinate frame remains attached to the center of mass and moves with the body. The advantage is that the inertial tensor is constant, unlike in the inertial frame where it varies with the configuration of the mechanism. However, expressing absolute accelerations in the body frame requires additional terms (e.g., $a_X = \dot{v}_X - \omega v_Y$). While body frames are generally preferred to have a constant inertial tensor, in the planar case, inertial frames can be used without any problems.

Connecting rigid bodies: joints. Going back to the formulation in the inertial frame, to interconnect rigid bodies and obtain complex mechanisms, it is necessary to define joints that constrain motion in specific directions. A **rotational joint** constrains the two connected bodies to have the same position (zero relative linear velocity) while leaving angular velocity free. Any difference in angular velocity ($\omega_r = \omega_B - \omega_A$) means a torque exists. A **translational joint** constrains relative motion to one direction. Since velocities are in the inertial frame, this direction is not fixed and is related to an angle ϕ. The relative velocities must be transformed into the direction of the joint. Both are represented by setting appropriate velocity-difference bonds to zero and using MTF elements for the direction transformation in the inertial frame. This is the basis for the bond graph modelling of robot manipulators and mechanisms.



3.5 Hydraulic and acoustic circuits

We now model some special but important classes of fluid-flow systems.

One category is **hydrostatic systems**. Typical examples are pumps, motors, pipes, pistons, valves, filters, and accumulators that use nearly incompressible fluids. Such systems are found in machine tools, earth-moving equipment, power transmissions, and aircraft control surface servomechanisms. These systems have high pressures and low fluid velocities so that the static pressure dominates the dynamic pressure. Pressure is effort, volume flow rate Q is flow.

A second category involves compressible gases where the pressure deviations are small enough that the **acoustic approximation** applies. In this regime, the gas behaves linearly, sound waves can be described using simplified wave equations, and we can avoid considering thermodynamic effects. Pneumatic systems with large pressure variations need a more complicated modeling treatment.

For both, the construction procedure follows the same 0/1-junction rules as electrical circuits:

1. Indicate positive directions for fluid flows and pressures.
2. For each distinct absolute pressure introduce a 0-junction. Attach elements if they react to one of the absolute pressures. If necessary, include a 0-junction for atmospheric pressure (which can be eliminated if "gauge" pressures are used).
3. Insert R-, C-, and I-elements between appropriate 0-junctions using 1-junctions that establish pressure differences acting on the passive elements.
4. Simplify the graph.

3.5.1 Fluid Resistance

A 1-port fluid resistor relates a pressure drop to a volume flow rate. The value of the resistance can be measured experimentally. A linear resistance can be assumed for a long, thin tube in which laminar flow (fluid flows in parallel layers, not turbulent) of an incompressible fluid develops: $R = 128\mu l/(\pi d^4)$. If the Reynolds number is low ($Re < 2000$), $P = RQ$. If it is turbulent ($Re > 5000$), $P = a_t Q|Q|^{3/4}$.

Short obstructions like orifices and valves are important cases where a pressure drop occurs over short lengths. For an orifice with a fixed area, the pressure drop is proportional to the square of the volume flow rate ($P = \rho Q|Q|/(2C_d^2 A_0^2)$), where the discharge coefficient C_d depends on the geometry and nonidealities. For valves, the area $A(x)$ depends on a position coordinate. The position of a mechanical part influences the hydraulic resistance, acting as a transducer linking the mechanical and hydrostatic domains. They can be modeled by variable-resistance elements modulated by a displacement $\dot{z}$.

3.5.2 Fluid Capacitance

A fluid capacitor imposes a relationship between a pressure and the integral of the volume flow rate (a volume of fluid). For a straight-walled water tank in a gravity field, $C = A/(\rho g)$ (linear). To model fluid compressibility in a rigid pipe, the bulk modulus B is used: $C = V_0/B$. For gases under the acoustic approximation, $C = V_0/(\rho_0 c^2)$. Sometimes capacitance is due to the elasticity of the vessel containing the fluid. For a thin-walled pipe, using the hoop-stress formula, $C = 2r_0 V_0/(t_w E)$. If both fluid compressibility and pipe elasticity contribute, an equivalent compliance can be constructed: $C_{eq} = V_0(1/B + 2r_0/t_w E)$.

When fluid compliance cannot be modeled with linear behavior, nonlinear models are essential. An example is a gas accumulator with a rubber bladder containing compressed gas. Using the isentropic pressure-volume law ($PV^\gamma = const$), the pressure is a nonlinear function of the volume: $P = P_0 V_0^\gamma/(V_0 - V)^\gamma$.

3.5.3 Fluid Inertia

Fluid systems have inertia effects analogous to mechanical systems: fluid elements store kinetic energy. The inertia coefficient is related to the parameters of a pipe segment. From Newton's law, a linear inertia coefficient for a pipe of constant cross-section A and length l is $I = \rho l/A$. For a pipe with a non-constant section, $I = \int_0^l \rho/A(s)ds$.

Watch out: Pseudo-bond graphs in hydraulics

In some hydraulic treatments, momentum $p_P = \int P\,dt$ and volume $V = \int Q\,dt$ are used as state variables, but the product of the chosen effort–flow pair may not equal power in non-standard formulations. Always verify that $e \cdot f$ has units of watts before coupling hydraulic submodels to other domains.

3.6 Transducers & multi-energy domain models

Transducers are devices that couple subsystems of distinct energy domains. A major reason for studying bond graphs is that they provide a uniform and precise way to represent systems when several forms of power and energy are involved.

Transformer transducers. An ideal transformer transducer, like a hydraulic ram (piston), converts mechanical power to hydraulic power: $F/A = P$ and $AV = Q$. This applies only to the ideal effects at the face of the piston. Real devices exhibit power losses and energy-storing mechanisms. The process consists of enriching the idealized model. On the mechanical side, we add a 1-junction with a resistor to include friction (which may be dry friction, $F_f = F_0 sgn(V)$, or linear, $F_f = bV$) and an I-element for the piston's mass. On the hydraulic side, a 0-junction and resistance represent leakage, and a C-element models fluid compliance (which may be nonlinear and depend on the nominal extension x_0). This complexity can lead to oscillatory phenomena that are damped by the resistance.

Positive displacement hydraulic pumps and motors consist of pistons moved by a shaft. With many pistons, the angular speed is related to the flow rate by a nearly constant coefficient: $P_B - P_A = T\tau$ and $TQ = \omega$. This is also modeled by a TF element.

Gyrator transducers. Electromechanical transducers, such as rotary/linear motors or voice coils, use gyrators. A current-carrying conductor moving in a magnetic field experiences an induced voltage (Faraday's law, $e = BlV$) and produces a force (Lorentz force, $Bli = F$). The cross-coupling of effort to flow and flow to effort makes this a gyrator ($e/V = GY = F/i$).

For a DC motor/generator with a commutator, the induced voltage is proportional to the angular speed, and the torque is proportional to the current: $e = T\omega$ and $Ti = \tau$. A detailed model is developed by adding the winding resistance and inductance, and the rotor's inertia and friction to the basic GY element.

Modulated Transducers. If the gain is time-dependent, MTF and MGY can model more complex power transducers. For example, an AC generator acts like a single piston/crank. The flux linkage depends on the angle θ, leading to $e = (nBl_2l_1 \cos\theta)\omega$ and $(nBl_2l_1 \cos\theta)i = \tau$. This is modeled by a Modulated Gyrator (MGY). Alternatively, using the Lorentz force and the lever law, it can be modeled as a GY followed by a Modulated Transformer (MTF). Similarly, the relation in a slider-crank mechanism between torque/angular speed and force/velocity is governed by a purely geometric transformation ($F\dot{x} = \tau\omega$ with $\dot{x} = f(\theta)\omega$). This is modeled as an ideal, power-conserving MTF with no inherent losses or energy storage.

The best model is the simplest one capable of demonstrating the aspects of the behavior that need to be understood. Extra loss and dynamic elements must be added with restraint.

Part II

Causality, Equations and Advanced Structures

Chapter 4

Causality and System Equations

In the world of state-space equations, the very same system can be described with different (but equivalent) sets of differential equations. The whole order of the system equations will always be the same (equal to the order of the model). For instance, a system can be represented by a single nth-order differential equation in terms of one unknown variable, or n first-order coupled differential equations in terms of n unknown variables (system theory approach), or various combinations of unknowns and equations of appropriate orders obtained with the Lagrangian approach.

We aim for a unified and easier methodology: Bond Graphs! One reason for constructing bond graphs is that it is straightforward to derive system equations and to develop numerical simulations. We can also go from a bond graph directly to a simulation without first obtaining the equations of motion manually. The real virtue of bond graph modelling becomes apparent when one considers that all system models use the same symbols. One formulation and solution procedure is needed!

Assigning causality to a bond graph is the step that transforms a structural description into a computational one. It decides which variables are inputs and which are outputs at every port, propagates these choices through the graph, and ultimately determines the mathematical character of the resulting model. The systematic procedure for doing this — the Sequential Causal Assignment Procedure (SCAP) — is the workhorse algorithm for deriving state-space equations from bond graphs.

4.1 The augmented bond graph

Before deriving equations, the bond graph must be **augmented**:

1. The bond graph has been designed, and an appropriate power convention has been assigned.
2. All bonds are numbered consecutively. This creates the *computational bond graph*.
3. Each bond is assigned a causal orientation (causal stroke) for the e and f variables.

The causal strokes record, for every bond, whether effort flows left-to-right or right-to-left — i.e. which element is computing each variable. The power half-arrow and the causal stroke are completely independent pieces of information. The half-arrow records the assumed positive power direction; the causal stroke records the computational input–output assignment. Do not confuse them. Power sign convention and causality assignment are totally unrelated.

4.2 Causality constraints by element type

In a causal sense, there are two types of bond graph elements: sources (Se and Sf) and junctions (0, 1, TF, and GY), that must meet certain causal conditions or their basic definitions are no longer valid. If a system cannot meet these causal conditions, the basic physical model on which the bond graph is based must be restudied; an impossible situation that is not capable of sensible mathematical resolution has been created.

Every element imposes constraints on the causalities of its bonds:

- **Se**: must impose effort — the causal stroke is always *away* from the Se.
- **Sf**: must impose flow — the causal stroke is always *toward* the Sf.
- **C (integral causality)**: flow is input, effort is output — stroke away from C. A capacitor in integral causality is used as a charge storage and for this it fills having its own voltage.

- **C (derivative causality)**: effort is input, flow is output — stroke toward C. This is the *non-preferred* form. A capacitor in derivative causality is used as a current generator to keep up lacks of voltage by discharging $I = C\dot{V}$.
- **I (integral causality)**: effort is input, flow is output — stroke toward I. An inductor in integral causality is used as a momentum storage and for this it fills having its own current (speed).
- **I (derivative causality)**: flow is input, effort is output — stroke away from I. Non-preferred. An inductor in derivative causality is used as a voltage generator to keep up lacks of current by discharging $V = L\dot{I}$.
- **R**: generally indifferent (either causality), unless the constitutive law is not invertible (e.g. Coulomb friction, which is not a bijection in e–f).
- **TF**: one port effort-in, the other effort-out (one stroke in, one out).
- **GY**: both ports effort-in, or both effort-out (both strokes on the same side).
- **0-junction**: exactly one bond has its effort directed *into* the junction (one stroke pointing away from the junction); all others have effort directed out (stroke toward junction).
- **1-junction**: exactly one bond has its flow directed *into* the junction (one stroke toward junction); all others have flow directed out (stroke away from junction).

These constraints mean that once any one element in a connected sub-graph gets its causality assigned, that assignment propagates through junctions and 2-port elements like a chain reaction. The constraint elements represent the physical structure in the system, and assigning causality to them means that they are used correctly in a particular "input-output fashion".

4.3 The Sequential Causal Assignment Procedure (SCAP)

Step 1. **Assign source causalities.** For each Se and Sf, assign the only admissible causal form. Propagate the implications through all constraint elements (0, 1, TF, GY) which may be connected to it.

Step 2. Repeat step 1 until all sources have been assigned the required causality.

Step 3. **Assign storage elements (preferred causality).** For each C and I element not yet causally assigned, assign *integral causality* (preferred). Propagate through constraint elements.

Step 4. Repeat step 3 until all storage elements have been assigned a causality.

Step 5. **Assign R-elements (if still needed).** If bonds remain unassigned after steps 1–4, choose any unassigned R-element and assign an arbitrary causality to it, and extend the causal implications through the graph. Each such arbitrary assignment signals an **algebraic loop**.

Step 6. Repeat step 5 until all R-elements have been used.

Step 7. **Assign remaining bonds arbitrarily.** Any remaining unassigned bond (joined to two constraint elements) can be assigned arbitrarily. Extend the causal implications. Each produces another algebraic loop.

Step 8. Repeat step 7 until all remaining bonds have been assigned.

The procedure is straightforward and orderly, and it can be implemented in a simulation environment. Note that in assigning causality, the results do not depend on the order of bonds chosen (except in special circumstances).

Key Insight: What the SCAP tells you

Several situations can arise when applying the SCAP, revealing the type of mathematical model:

- **All storage elements in integral causality, and the graph is complete after step 4**: the simplest and most common case. The model yields explicit ODEs directly.
- **Causality is completed using steps 5-8**: one or more *algebraic loops* exist. The model involves coupled algebraic–differential equations; some effort or flow variable is algebraically related to itself and must be solved simultaneously.

- **Some storage elements forced into derivative causality (during step 3)**: the model contains *implicit* differential equations (DAEs). The number of derivative-causality storage elements equals the *reduction in system order* relative to the total element count.

4.4 Reading off state-space equations: Basic formulation & Reduction

From a fully augmented bond graph model, the equations for the system can be developed in a very orderly fashion, either by hand or thanks to a simulation package. Very general and powerful procedures are available for producing sets of system equations, consisting of these main steps: select input and energy state variables, formulate the initial set of system equations, and reduce the initial equations to state-space form.

The selection of inputs is straightforward, since they are associated to each source element. These variables will appear in the final state-space equations if they have any effect on system behaviour.

The state variables are:

- The **momentum** p on every I-element in *integral* causality.
- The **displacement** q on every C-element in *integral* causality.

It is useful to indicate the **co-energy variables**: flows on the I-elements ($f = \Phi_I^{-1}(p)$) and efforts on the C-elements ($e = \Phi_C^{-1}(q)$). Co-energy variables appear in the initial formulation and then they are eliminated during the reduction process since they are function of the state.

The order of the system equals the number of energy storage elements in integral causality. The derivation procedure is systematic:

(1) For each state variable, write its rate of change (time derivative) as an effort or a flow: $\dot{p} = e$ or $\dot{q} = f$.

(2) Causality is used to track the efforts and flows through the bond graph. The causal stroke tells us which element is *providing* this effort or flow. Trace through the bond graph from the state variable's bond, substituting at each junction and element, until only sources and **co-energy variables** remain. It makes no difference which state variable we start with.

(3) Substitute the co-energy variables last to obtain equations purely in state variables and inputs. This leads to explicit first-order equations.

■ *Example 1: Formulation of a simple 2-state system*

See slides 18-19, pack M04

Derivation: Simple example: RC circuit

State variable: q_2 on C. Input: $e_1(t)$ from Se.

From the 0-junction: $e_2 = e_1 - e_4$ (0-jct, effort balance).
From the 1-junction: $e_4 = R_4 f_4 = R_4 f_2$ (R-element, effort-in causality, same flow on 1-jct).
Co-energy: $f_2 = \dot{q}_2$, and $e_2 = q_2/C_2$.

Result: $\dot{q}_2 = f_2 = \frac{e_1 - q_2/C_2}{R_4}$ — the standard RC first-order equation.

The 1-junction encodes KVL, the 0-junction encodes KCL. Reading equations from a bond graph is essentially applying Kirchhoff's laws automatically: the 0-junction says "efforts sum to zero around a node" (I Kirchhoff); the 1-junction says "flows sum to zero around a loop" (V Kirchhoff). Causality tells you which variable is expressed in terms of the others.

■ *Example 3: System with a transformer and multiple states*

See slides 20-23, pack M04

4.5 Algebraic loops

Talking about state-space equations formulation, in situations in which we do NOT have algebraic-loops and we have ALL storage energy elements in integral causality, the formulation is very straight-forward, obtaining explicit differential equations. Instead, when we have algebraic loops (we have encountered steps 5-6 of the SCAP) and/or we have some energy storage elements in derivative causality, the formulation is not anymore so immediate!

An algebraic loop arises when a chain of elements forms a closed ring with no energy-storage element breaking it. In that loop, one variable (effort or flow) becomes algebraically related to itself (as well as to other inputs and co-energy state variables). Tracing around the loop brings you back to your starting point: $e_1 = F_1(e_2, f_2)$, $e_2 = F_2(e_3, f_3) \dots$ until we reach again the initial variable.

The procedure for handling algebraic loops is to first make an arbitrary causal assignment on one of the causally unassigned bonds and then propagate the causal information as far as possible. If the causality completes after one arbitrary assignment, then there is one algebraic loop. The more loops involved, the more difficult the final formulation becomes.

Watch out: Cause of algebraic loops

Algebraic loops always arise from a **modelling simplification**. For example, two parallel dampers with no masses to separate them create an algebraic loop in force. In reality, there would be some small compliance or inertia in between; the idealisation of perfectly rigid connections removes it and creates the loop.

When SCAP is stuck on step 4, it is necessary to arbitrarily assign causality to one bond. It is recommended that when a resistance element is involved in the algebraic loop, the resistance element be chosen for the arbitrary assignment (step 5). It is customary to choose the variable that was arbitrarily selected to be the output from the R-element and the input to the rest of the system.

In order to solve the algebraic loop, we have to collect all the equations involved in the loop and solve the system to find an expression of e/f that depends *only* on inputs and/or co-energy state variables. This explicit algebraic equation can then be substituted directly into the ODEs.

For linear systems, this procedure will work for multiple loops, although solving by hand becomes formidable rather quickly. For nonlinear systems, it may not be possible to invert some of the algebraic relationships (e.g. Coulomb friction). When these inversions appear to be impossible, we HAVE TO eradicate the problem at the root by passing to less simplistic modelling assumptions!

■ *Algebraic Loop Example: Assigning arbitrary causality on an R-element*

See slides 26-29, pack M04

4.6 Derivative causality

Another possible consequence of certain modelling assumptions is that, after sequential causal assignment, one or more energy storage elements are forced into derivative causality. When a C-element is forced to accept derivative causality (effort-in), it means that its effort (and hence its displacement $q = Ce$) is

algebraically determined by the rest of the system — it does not contribute an independent state variable to the final equations. This occurs when two C-elements are connected in a configuration that rigidly constrains their displacements (e.g. two springs rigidly connected without an intervening mass). The system order is reduced by one per derivative-causality C.

Similarly for I-elements in derivative causality (flow-in): their momenta are algebraically determined. The independent state variables remain the energy variables in the energy storage elements with integral causality. (Note: The derivative elements do still store energy!).

The true problem arises when formulating the equations. When we encounter a C/I element in derivative causality, we are guaranteed to end up with its power variable that is NOT the co-energy one ($\dot{q}$ for C-elements and $\dot{p}$ for I-elements). We end up with a double problem: we end up with the derivative of an energy variable, AND this energy variable is NOT a state one!

These situations always signal an **implicit (DAE) model**. Implicit state equations can be algebraically transformed to explicit ones, but this is not necessarily a simple task.

Handling Derivative Causality Algebraically: Given the dependent energy variable of a C/I element in derivative causality, it is always guaranteed that it is algebraically related (constrained) to variables that are independent state ones.

1. Find out (by inspecting the BG) the algebraic relation that gives the energy dependent variable as a function of the state ones.
2. Differentiate that relation to obtain an expression for $\dot{q}$ (for C) or $\dot{p}$ (for I) as a function of differentiated state variables.
3. Substitute this back into your equations. The result is a valid state equation, but it will be in implicit form.
4. To get it in explicit form, further algebraic manipulations are required to isolate the derivatives of the state variables on the left side.

■ *Derivative Causality Example: Neglecting motor inertia*

See slides 34-36, pack M04

Handling Derivative Causality at the Root: Derivative causality is ALWAYS caused by too simplistic modelling assumptions! It physically indicates an over-idealisation (e.g. perfectly rigid connections between compliant elements, or massless inertial elements).

It is always possible to revisit the modelling assumptions that resulted in the formulation problem and change these assumptions to yield a model that is virtually the same as the original one, but without the formulation problems. Introducing a new energy storage element (a tiny mass or compliance) restores integral causality and ordinary ODE structure.

There are some computational penalties that accompany very small added inertias or very stiff added compliances in a model (it introduces a fast time scale, making the system "stiff" for numerical solvers). However, for some systems, these penalties are a very small price to pay for eliminating formulation problems at the root.

4.7 System order and physical meaning

The order of a bond graph model equals the number of independent energy storage elements. This is a fundamental invariant: it does not matter which set of state variables is chosen (positions vs. momenta, etc.) — the order is always the same. Equivalently, the order equals the minimum number of initial conditions needed to uniquely determine future evolution.

The energy of the system at any instant is the sum of the energies stored in all elements with integral causality:

$$E_{\text{total}}(t) = \sum_{C\text{-elements}} E_C(q) + \sum_{I\text{-elements}} E_I(p). \tag{4.1}$$

This is the **Hamiltonian** of the system (in the linear case, $E = \frac{1}{2}\mathbf{x}^T Q\mathbf{x}$ for an appropriate positive-definite matrix Q).

Chapter 5

Multiport Fields and Junction Structures

Real physical components often couple multiple energy flows simultaneously. A flexible beam deformed by two forces stores energy that depends on *both* displacements simultaneously, not independently. A 3D rigid body has coupled translational and rotational inertia. These multi-port couplings cannot be captured by separate 1-port elements — they require **multiport fields**.

Using fields and junction structures, it is possible to study systems with complex multiport components using bond graphs because they combine structural detail, clarity, and ease of visualisation.

Key Insight: Important Assumption for Fields

All multiport elements that we are going to study in this chapter are assumed to be (and are going to be studied as) **ALL INWARDS** in terms of power bond directions. This means power is conventionally positive when flowing into the field from all ports.

5.1 C-fields: multiport potential energy storage

A **C-field** is the multiport generalisation of the 1-port C-element. It is itself a distributed compliance element, representing a distributed hole of energy conservation. It has n ports, and its potential energy is a function of all n displacements simultaneously:

$$E(\mathbf{q}) = \int_{\mathbf{q}_0}^{\mathbf{q}} \mathbf{e}(\bar{\mathbf{q}})^T \, d\bar{\mathbf{q}}, \qquad e_i = \frac{\partial E}{\partial q_i}(\mathbf{q}). \tag{5.1}$$

This means that a change in any component is going to cause a change in the stored energy. Because $E(\mathbf{q})$ represents stored potential energy, the C-field inherently conserves energy.

The constitutive law is a set of n functions $e_i(\mathbf{q})$ giving the efforts at each port as a function of all displacements. By inspection, $\mathbf{e}(\mathbf{q}) = \nabla E(\mathbf{q})$. For the C-field to be physically meaningful (conservative), the energy function $E(\mathbf{q})$ must exist, which requires the **Maxwell reciprocal relations** (a structural constraint):

$$\frac{\partial e_i}{\partial q_j} = \frac{\partial^2 E}{\partial q_i \partial q_j} = \frac{\partial e_j}{\partial q_i} \quad \forall\, i, j. \tag{5.2}$$

This is the direct multivariable generalisation of the single-port condition $e = \partial E/\partial q$. It is clear that not every arbitrary set of constitutive laws expressing efforts in terms of displacements could come from a valid energy-storing C-field—only those sets that obey this symmetry property!

Compliance and stiffness forms. If the constitutive law can be written $\mathbf{q} = \mathbf{M}_c(\mathbf{e})$ (displacements as functions of efforts), this is the **compliance form**. If it can be written $\mathbf{e} = \mathbf{M}_k(\mathbf{q})$ (or $\mathbf{M}_c^{-1}(\mathbf{q})$), this is the **stiffness form**.

In the linear case, the equations become:

$$\mathbf{e} = \mathbf{K}\,\mathbf{q} \quad \text{(stiffness)}, \qquad \mathbf{q} = \mathbf{C}\,\mathbf{e} \quad \text{(compliance)}, \tag{5.3}$$

where $\mathbf{K} = \mathbf{C}^{-1}$ is the stiffness matrix and $\mathbf{C} = \mathbf{M}_c$ is the compliance matrix. The stored energy is $E = \frac{1}{2}\mathbf{q}^T\mathbf{K}\mathbf{q}$. Maxwell reciprocity requires $\mathbf{K} = \mathbf{K}^T$ and $\mathbf{C} = \mathbf{C}^T$ (symmetric matrices), a familiar result from structural mechanics.

Example 5.1 (Elastic beam loaded at two points)**.** An Euler-Bernoulli beam of length L, stiffness EI, loaded by forces F_1 at $L/3$ and F_2 at $2L/3$ has the compliance matrix:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{L^3}{243EI} \begin{pmatrix} 4 & \frac{7}{2} \\ \frac{7}{2} & 4 \end{pmatrix} \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}. \tag{5.4}$$

The symmetric compliance matrix confirms Maxwell reciprocity.



Nonlinear C-fields. Nonlinear C-fields are common in certain application areas. A mechanical element that behaves as a linear 1-port in restricted cases can often be modelled as a nonlinear 2-port C-field when two-dimensional motion is permitted. For example, a simple linear spring (rest length L_0, stiffness k) pivoting in a 2D plane (x, y) has the energy:

$$E(x, y) = \frac{1}{2}k\left(\sqrt{x^2 + y^2} - L_0\right)^2 \tag{5.5}$$

The gradient of this potential energy gives the highly nonlinear constitutive equations for the x and y efforts:

$$F_x = kx - k\frac{L_0 x}{\sqrt{x^2 + y^2}}, \qquad F_y = ky - k\frac{L_0 y}{\sqrt{x^2 + y^2}} \tag{5.6}$$



5.1.1 Explicit vs. implicit C-fields

An **explicit field** is one whose constitutive law is given directly as $\mathbf{e}(\mathbf{q})$ or $\mathbf{q}(\mathbf{e})$ — the multiport is defined from the outset as a single entity (like the beam example above).

An **implicit field** arises when a group of 1-port C-elements, connected by junction structures (0-junctions, 1-junctions, TF), collectively behaves as a C-field. For example, two springs k_1 and k_2 connected in series via a 1-junction (sharing a common force, but adding displacements) or connected in parallel via a 0-junction form an implicit C-field. The implicit form is naturally obtained from the assembly of the bond graph; converting it to explicit form requires solving for the field's constitutive equations algebraically.

Key Insight: When to convert implicit to explicit

Converting an implicit field to an explicit form (Viva Explicit C-Fields!) is extremely useful when: (a) Some elements of the field are forced into derivative causality. By pre-solving the algebraic loops and constraints inside a field matrix, you save the computer from crashing during simulation and kill the derivative causality mathematically. (b) Equation formulation is simplified by working directly with the field's stiffness or compliance. The transformation into an explicit form integrates trivial state equations once and for all, avoiding the necessity of handling derivative causality completely.

5.1.2 Causality in C-fields

As in the case of 1-port C-elements, multiport C-fields also admit mixed integral-derivative causalities. A C-field with n ports can accept any mixture of integral and derivative causalities across its ports, as long as the resulting constitutive equations can be inverted appropriately.

In **fully integral causality**, all flows are inputs and all efforts are outputs: $\mathbf{e} = \mathbf{K}\,\mathbf{q}$ (stiffness form). In **fully derivative causality**, all efforts are inputs and all displacements are outputs: $\mathbf{q} = \mathbf{C}\,\mathbf{e}$ (compliance form). **Mixed causality** involves both, requiring a mixed-form constitutive law obtained by partially inverting the stiffness/compliance matrix.

For an implicit field, admissible causal forms are easily determined by propagating the standard bond graph causality rules through its constituent elements (e.g., studying conflicts at 0 or 1 junctions). For an explicit field specified in a particular causal form, the question is an algebraic one (whether the constitutive equations can be inverted), and for nonlinear fields, this can be very hard to decide. Notice that, in principle, any strictly valid C-field can always go into fully integral causality!

5.2 I-fields: multiport kinetic energy storage

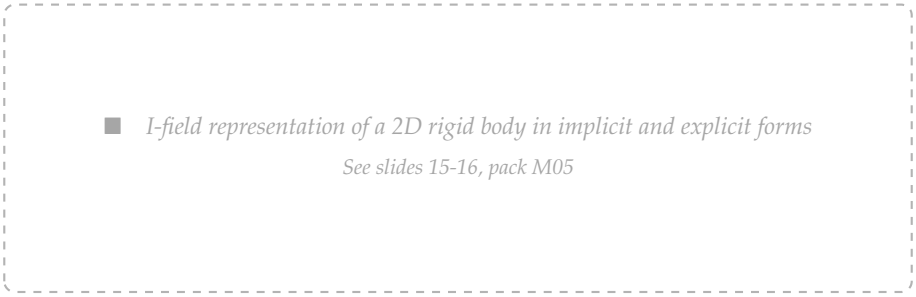
For I-fields we can develop a complete parallel dual formulation with respect to C-fields. Inertial fields are the exact duals of capacitive fields. The **I-field** is the multiport generalisation of the 1-port I-element. Its kinetic energy is a function of all momenta simultaneously:

$$E(\mathbf{p}) = \int_{\mathbf{p}_0}^{\mathbf{p}} \mathbf{f}(\bar{\mathbf{p}})^T \, d\bar{\mathbf{p}} \quad \longrightarrow \quad f_i = \frac{\partial E}{\partial p_i}(\mathbf{p}) \tag{5.7}$$

In the linear case, the energy is $E(\mathbf{p}) = \frac{1}{2}\mathbf{p}^T\mathbf{M}^{-1}\mathbf{p}$. The constitutive law maps momenta to flows via the inverse mass/inertia matrix $\mathbf{M}^{-1}$ (i.e. $\mathbf{f} = \mathbf{M}^{-1}\mathbf{p}$).

The Maxwell reciprocal relations for I-fields require $\partial f_i/\partial p_j = \partial f_j/\partial p_i$, which translates to the symmetry of $\mathbf{M}^{-1}$ (and hence of $\mathbf{M}$). Like C-fields, I-fields occur in both implicit and explicit forms, and manipulating them can yield highly useful simplified representations.

The most important mechanical physical I-field is the **3D rigid body**: it has three translational and three rotational momentum ports, with the (constant) mass-inertia tensor providing the constitutive law. In 2D, the I-field reduces to two translational inertias and one rotational inertia, as discussed in Chapter 3. By transforming an implicit rigid-body I-field into an explicit one, we can define the relationship between the center-of-mass velocity and the velocities of the attachment points analytically.



Interacting Coils (Mutual Inductance). Another classic use of I-fields is in electrical circuits containing mutually interacting coils. These are conveniently represented using explicit I-fields in derivative causality. To write the constitutive laws, one needs a sign convention for currents/voltages and a convention for the relative orientation of the coils (often denoted by dots, squares, or triangles on schematic diagrams).

If the currents i_1 and i_2 are defined so that when positive they both enter the dotted ends, the mutual-inductance effects M are positive:

$$\begin{pmatrix}\lambda_1\\ \lambda_2\end{pmatrix} = \begin{pmatrix}L_1 & M_{12}\\ M_{12} & L_2\end{pmatrix}\begin{pmatrix}i_1\\ i_2\end{pmatrix} \tag{5.8}$$

If one current enters a dot and the other leaves a dot, the mutual terms become $-M_{12}$. Independently of the coil orientation convention, the inductance matrix and its inverse will always be symmetric as long as the I-field is conservative.



5.3 Mixed energy-storing fields (IC-fields)

There are cases when an energy-storing device acts as a C-field at some ports and an I-field at others. Within a single energy domain, there is usually no need for such an element. However, energy transducers are often ideally energy conservative but store and release energy across different domains over time (e.g., storing electrical energy and releasing it as mechanical work). Because their electrical and mechanical energies are dynamically coupled, they cannot be modeled by simple 1-port C or I elements. You must use a mixed **IC-field** to mathematically capture the complex physics natively.

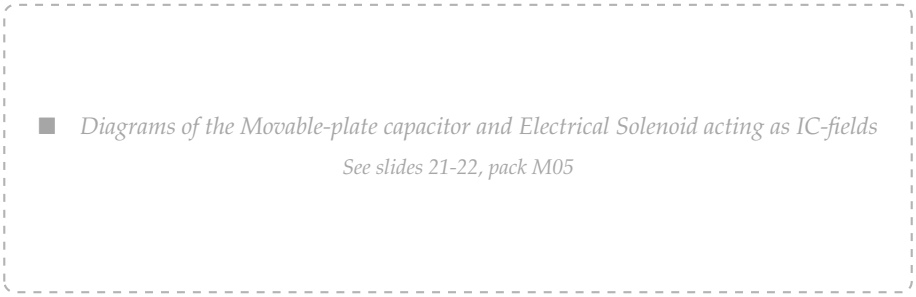
For an IC-field with momentum variables p and displacement variables q, the total energy features dynamic coupling (an irrotational case) and contains mixed terms. The constitutive equations are obtained by taking the appropriate partial derivatives of the total stored energy $E(p, q)$.

Example 5.2 (Movable-Plate Capacitor)**.** A capacitor where one plate can move mechanically acts as a C-port electrically (charge q) and a C-port mechanically (position x). The stored energy is $E(q, x) = \frac{1}{2}\frac{q^2}{C(x)}$. The constitutive equations are:

$$e = \frac{\partial E}{\partial q} = \frac{q}{C(x)} \qquad \text{and} \qquad F = \frac{\partial E}{\partial x} = -\frac{1}{2}\frac{q^2}{C^2(x)}\frac{\partial C}{\partial x} \tag{5.9}$$

Example 5.3 (Electrical Solenoid)**.** The electrical solenoid serves as a prototype of an IC-field transducer. From the electrical port, the coil exhibits self-inductance (λ). From the mechanical port, the force on the slug depends on its position (x). The energy is $E(\lambda, x) = \frac{1}{2}\frac{\lambda^2}{L(x)}$. The constitutive equations are:

$$i = \frac{\partial E}{\partial \lambda} = \frac{\lambda}{L(x)} \qquad \text{and} \qquad F = \frac{\partial E}{\partial x} = -\frac{1}{2}\frac{\lambda^2}{L^2(x)}\frac{\partial L}{\partial x} \tag{5.10}$$



5.4 R-fields: multiport dissipation

An **R-field** is an n-port with a static (or algebraic) constitutive law that relates efforts and flows. In general, R-fields are **NOT** strictly related or associated to power losses: their TRUE intrinsic characteristic is to algebraically relate efforts and flows!

This broad definition includes power-conserving elements, such as 0- and 1-junctions, and elements containing sources. However, in practice, most explicit R-fields do dissipate power (e.g. thermal resistors, coupled fluid friction matrices). Explicit R-fields arise frequently in modelling nonlinear devices.

The causality of an R-field is determined by the sources and energy-storing elements attached to it.

- **Resistance causality:** All flows are inputs, efforts are outputs ($e_i = \Phi_{R_i}(f_1 \dots f_n)$).
- **Conductance causality:** All efforts are inputs, flows are outputs ($f_i = \Phi_{R_i}^{-1}(e_1 \dots e_n)$).
- **Mixed causality:** A mixture of effort and flow inputs.

Onsager vs. Casimir Forms and Power. Since an R-field has no stored energy function, there is no direct equivalent to Maxwell's Reciprocal Relations. However, structural properties do exist. Any square Constitutive Law Matrix (CLM) M can always be mathematically decomposed into a symmetric part $M_S = \frac{1}{2}(M + M^T)$ and a skew-symmetric (antisymmetric) part $M_A = \frac{1}{2}(M - M^T)$.

The asymmetric contribution M_A is always related to *perfectly conserved power* ($P_w = b^T M_A b = 0$). The symmetric part M_S is associated with power dissipation.

Implicit R-fields composed strictly of linear 1-port resistances, 0-junctions, 1-junctions, and transformers will obey **Onsager Reciprocity**: when expressed in pure resistance or conductance form, their CLM is perfectly symmetric (called the *Onsager form*). When expressed in mixed causality, the matrix is guaranteed to present asymmetric terms.

Importantly, NOT all R-fields obey Onsager Reciprocity. If Gyrators (GY) are allowed inside the implicit R-field, the reciprocity is broken, yielding asymmetric terms even in pure resistance/conductance forms. This is known as the *Casimir form*.

5.5 Modulated 2-port elements

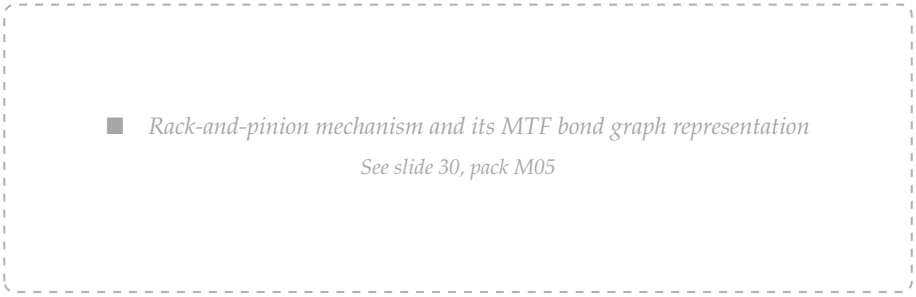
Before discussing pure junction structures, it is essential to introduce the modulated 2-ports. The **modulated transformer (MTF)** and **modulated gyrator (MGY)** are nonlinear, power-conserving generalisations of TF and GY.

The parameters (modulus m or r) of these elements are allowed to be a function of some other parameter in the system (often a displacement or angle, possibly time-varying). These parameters are modulated by means of a signal (active bond) rather than a power bond. Thus, it is characteristic of these elements that the parameters change value without a directly associated power flow!

Examples include the DC motor (Faraday & Lorentz laws modelled as an MGY depending on field current) and the slider-crank or rack-and-pinion mechanisms (modelled as MTF).

Watch out: Rack-and-Pinion Oversimplification

A rack-and-pinion might look like a simple MTF relating linear velocity to angular velocity ($V_2 = r\omega$) and forces ($F_1 = rF_2$). However, if you strictly model it this way, you assume the gear itself has no inertia and requires no power to rotate! Always pay attention to possible errors due to over-simplification.



5.6 Junction structures

Junction structures are the energy switchyards that enforce the constraints among parts of dynamic systems. They result from the interconnection of 0-junctions, 1-junctions, TF, and GY. They contain no dissipative or storage elements.

Junction structures are essentially special types of R-fields that *never* dissipate power (meaning their CLM is completely skew-symmetric). Because they cannot dissipate power, they typically **cannot** accept all possible causal assignments at their ports (e.g., a pure junction structure without gyrators cannot accept resistance or conductance causality on all ports simultaneously).

Junction structures have a key property: they are themselves power-conserving, so $\sum_k e_k f_k = 0$ across all their external ports. Equivalently, the junction structure defines a set of linear constraints (from the junctions) and scaling/exchange rules (from TF and GY) on the port variables.

Key Insight: Bond graphs as Dirac structures

From an advanced viewpoint, the junction structure of a bond graph defines a **Dirac structure** — a geometric object in power-port space that simultaneously generalises Kirchhoff's laws, kinematic constraints, and power-conserving transformations. This connection to geometric mechanics (port-Hamiltonian systems) is the theoretical foundation for structure-preserving simulation and passivity-based control.

5.6.1 Multiport Transformers (Kinematics)

Another extremely interesting viewpoint about junction structures involves the transformations of variables they provide. A **multiport transformer** is a specific assemblage of 0s, 1s, and TFs that shows more than two ports to the external world.

Mathematically, a multiport transformer is characterised by a matrix $\mathbf{G}$ and its transpose $\mathbf{G}^T$, just as a 2-port transformer is characterised by a single modulus m. By setting a "through" power convention (power in for inputs, out for outputs), the constitutive law can always be written as:

$$\mathbf{e}_{in} = \mathbf{G} \cdot \mathbf{e}_{out} \qquad \text{and} \qquad \mathbf{f}_{out} = \mathbf{G}^T \cdot \mathbf{f}_{in} \tag{5.11}$$

This is exactly the formulation used for 3D Kinematics and coordinate transformations! A typical example is implementing a Jacobian $J(q)$ that links a joint-coordinate space to a Cartesian-coordinate space in robotics. The forward kinematics ($\dot{x} = J(q)\dot{q}$) dictates the flow transformations, and by power conservation, the transpose Jacobian dictates the force-torque effort transformations ($T = J^T(q)F$). Grouping these into a multiport MTF field elegantly handles 3D kinematics directly within the bond graph framework.

5.7 IC-fields and Lagrangian mechanics

When a junction structure connects multiple I and C fields, the resulting overall structure can be equivalently described by Lagrange's equations of motion. Specifically, if the junction structure provides the kinematic constraints of a mechanism, and the I and C fields provide inertial and compliant energies, then the full bond graph model is equivalent to a Lagrangian description with generalised coordinates.

The **IC-field** is the combined multi-port storage element that results from this combination. Its energy is the total Lagrangian $L = T(\mathbf{p}) - V(\mathbf{q})$, and its constitutive equations are Hamilton's equations. This connection is central to the bond graph modelling of multi-body systems (Chapter 6).

Part III

Mechanical, Distributed and Thermo-fluid Systems

Chapter 6

Multi-body Systems

Extending bond graph modelling from 2D plane motion to full **3D rigid-body mechanics** brings geometric nonlinearity through the rotation group. These geometric nonlinearities are a primary cause of equation formulation problems. The key modelling choice — whether to express quantities in the inertial frame or a body-fixed frame — has direct consequences for the structure of the equations and the bond graph.

6.1 Dynamics of a 3D rigid body

Consider a rigid body of mass m with moments of inertia J_x, J_y, J_z along the principal axes. The body-fixed coordinate frame $\{G; x, y, z\}$ is attached to the centre of mass G with axes aligned with the principal axes of inertia.

Why the body frame? In the inertial frame, the inertia tensor $\mathbf{J}$ is a 3×3 matrix that varies with the body's orientation — making the constitutive equations of the I-field time-varying and complex. In the body frame, $\mathbf{J} = \mathrm{diag}(J_x, J_y, J_z)$ is **constant**, which is the primary advantage of this formulation. The mass m and inertia $\mathbf{J}$ are invariant, making them highly practical coordinates for computation.

The price is that Newton's law in the body frame contains **nonlinear gyroscopic terms** from the frame's own rotation. Newton's law applied in the body frame requires accounting for the rate of change of momentum relative to the moving frame.

Newton–Euler equations. Let $\mathbf{v} = (v_x, v_y, v_z)^T$ and $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)^T$ be the translational velocity of G and the angular velocity, *both expressed in the body frame*. Let $\mathbf{F}$ and $\boldsymbol{\tau}$ be force and torque at G in the body frame. Then:

$$\mathbf{F} = \dot{\mathbf{p}} + \boldsymbol{\omega} \times \mathbf{p}, \qquad \mathbf{p} = m\mathbf{v}, \tag{6.1}$$

$$\boldsymbol{\tau} = \dot{\mathbf{h}} + \boldsymbol{\omega} \times \mathbf{h}, \qquad \mathbf{h} = \mathbf{J}\boldsymbol{\omega}. \tag{6.2}$$

Written component-wise, these are the **Euler equations**:

$$\begin{aligned} F_x &= m\dot{v}_x + m\omega_y v_z - m\omega_z v_y \\ F_y &= m\dot{v}_y + m\omega_z v_x - m\omega_x v_z \\ F_z &= m\dot{v}_z + m\omega_x v_y - m\omega_y v_x \\ \tau_x &= J_x\dot{\omega}_x + \omega_y J_z\omega_z - \omega_z J_y\omega_y \\ \tau_y &= J_y\dot{\omega}_y + \omega_z J_x\omega_x - \omega_x J_z\omega_z \\ \tau_z &= J_z\dot{\omega}_z + \omega_x J_y\omega_y - \omega_y J_x\omega_x. \end{aligned} \tag{6.3}$$

These nonlinear differential equations have no general analytical solution. Furthermore, they provide velocities with respect to a frame that is aiming in a different direction at every instant. The cross-product terms $\boldsymbol{\omega} \times \mathbf{p}$ and $\boldsymbol{\omega} \times \mathbf{h}$ are gyroscopic: they do not do work (they are perpendicular to velocity), but they redistribute momentum between components. In the bond graph, they appear as **modulated gyrators** (MGY elements) whose moduli depend on the angular velocity — making the model intrinsically nonlinear.

Key Insight: The I-field of a 3D rigid body vs. Independent Elements

The 3D rigid body is conceptually represented by a **6-port I-field** in the body frame: three translational ports and three rotational ports. The constitutive law (flow = $\mathbf{M}^{-1}\mathbf{p}$, with block-diagonal $\mathbf{M} = \mathrm{diag}(m\mathbf{I}_3, \mathbf{J})$) is linear and constant.

However, *attention must be paid to the actual bond graph implementation*. In practice, this block is modelled as **six independent 1-port I-elements** (since the mass matrix is diagonal in the principal body frame), interconnected by an Eulerian junction structure containing the MGYs. The nonlinear gyroscopic interaction is handled by MGY elements in the junction structure, not inside the I-elements themselves. This separation of linear storage from nonlinear routing is elegant and computationally efficient. As long as external forces and torques are causal inputs, integral causality exists for all six I-elements.



6.2 Rotation matrices and coordinate transformations

Since all velocities and forces are expressed in the body frame, but external loads and gravity often come from the inertial frame (where e.g., gravity acts strictly downwards), coordinate transformations are necessary. We need an appropriate "junction element" to bridge the body frame and the inertial frame.

The rotation matrix $\mathbf{R}(t)$ (a 3×3 orthogonal matrix with $\mathbf{R}^T\mathbf{R} = \mathbf{I}$, $\det \mathbf{R} = 1$) transforms vectors from the body frame to the inertial frame ($\mathbf{w}_{inertial} = \mathbf{R}\mathbf{w}_{body}$). In bond graphs, $\mathbf{R}$ acts as a Modulated Transformer (MTF) that conserves power.

Yaw, Pitch, and Roll (Cardan Angles). Any spatial transformation consists of a series of coordinate rotations. For spinning bodies, Euler angles are typical, but for multi-body systems like robotic arms or aircraft, Yaw (ψ), Pitch (θ), and Roll (ϕ) (Cardan Angles) are frequently used. The combined rotation matrix is $\mathbf{R}_{\psi\theta\phi} = \mathbf{R}_\psi \mathbf{R}_\theta \mathbf{R}_\phi$.

By mapping the body-frame angular velocity $\boldsymbol{\omega}$ to the angle derivatives, we obtain the kinematic relationships:

$$\dot{\psi} = \frac{\sin\phi}{\cos\theta}\omega_y + \frac{\cos\phi}{\cos\theta}\omega_z \tag{6.4}$$

$$\dot{\theta} = \cos\phi\,\omega_y - \sin\phi\,\omega_z \tag{6.5}$$

$$\dot{\phi} = \omega_x + \sin\phi\frac{\sin\theta}{\cos\theta}\omega_y + \cos\phi\frac{\sin\theta}{\cos\theta}\omega_z \tag{6.6}$$

Care must be taken regarding kinematic singularities (e.g., when $\cos\theta = 0$). Alternatively, one can directly integrate the differential equation of the rotation matrix:

$$\dot{\mathbf{R}} = \mathbf{R}\,\mathbf{S}(\boldsymbol{\omega}) \tag{6.7}$$

where $\mathbf{S}(\boldsymbol{\omega})$ is the skew-symmetric matrix:

$$\mathbf{S}(\boldsymbol{\omega}) = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}. \tag{6.8}$$

The quantity $\dot{\mathbf{R}}\mathbf{R}^T = \mathbf{R}\,\mathbf{S}(\boldsymbol{\omega})\,\mathbf{R}^T$ is the angular velocity in the inertial frame expressed as a skew-symmetric matrix. In the bond graph, MTF elements with moduli from $\mathbf{R}$ handle these transformations when coupling to the environment or to other bodies.

Forces not applied at the Centre of Gravity. Often, connecting forces are applied at a point P offset from the CoG G by a fixed vector $\mathbf{r}_{GP}$ in the body frame. The velocity of point P is given by:

$$\mathbf{v}_P = \mathbf{v}_G + \boldsymbol{\omega} \times \mathbf{r}_{GP} \tag{6.9}$$

In matrix form, this kinematic relationship defines a **Body Frame Configuration Tensor** $\mathbf{T}(P)$:

$$\begin{pmatrix} \mathbf{v}_P \\ \boldsymbol{\omega} \end{pmatrix} = \begin{pmatrix} \mathbf{I} & -\mathbf{r}_{GP}\times \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{v}_G \\ \boldsymbol{\omega} \end{pmatrix} \tag{6.10}$$

By power conservation, the transposed tensor maps the forces applied at P back to equivalent forces and torques at the CoG:

$$\begin{pmatrix} \mathbf{F}_G \\ \boldsymbol{\tau}_G \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{r}_{GP}\times & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{F}_P \\ \boldsymbol{\tau}_P \end{pmatrix} \tag{6.11}$$

This allows us to freely attach any number of forces or velocities applied at generic points using an MTF block.

Body Frame Configuration Tensor MTF connecting offset forces to the Euler Equations block

See slides 12–15, pack M06

6.3 Joints between rigid bodies

A **joint** connects two rigid bodies by constraining their relative motion to a subspace. In the inertial frame:

Rotational (Revolute) joint. The relative linear velocity between the two attachment points is zero; the relative angular velocity $\omega_r = \omega_B - \omega_A$ is free (the joint can rotate). A torque τ_r acts along the joint axis. A revolute joint fully constrains linear velocities at the joint centre P, which is fixed with respect to both body frames. The bond graph imposes the velocity constraint via an MTF ($\mathbf{v}_P^A = \mathbf{R}_B^A \mathbf{v}_P^B$), and identifies the relative angular velocity:

$$\mathbf{v}_A = \mathbf{v}_B \quad \text{(at the joint location)}, \qquad \boldsymbol{\omega}_{AB}^A = \boldsymbol{\omega}_B^A - \boldsymbol{\omega}_A^A \text{ (free)}. \tag{6.12}$$

Assuming the z-axis is oriented along the rotation axis, $\boldsymbol{\omega}_{AB}^A = (0, 0, \dot{\varphi})^T$.

Translational (Prismatic) joint. A prismatic joint fully constrains angular velocities among the bodies ($\boldsymbol{\omega}_A = \boldsymbol{\omega}_B$), meaning they always have the same relative orientation. The relative linear velocity is constrained to one direction $\hat{\mathbf{d}}(\theta)$ (which itself varies as the mechanism rotates). A force F_r acts along this direction. Assuming the translation axis coincides with the z-axis of both frames, $\mathbf{v}_{AB}^A = (0, 0, \dot{s})^T$. The constraint and the direction transformation involve MTF elements with time-varying moduli that depend on the current configuration angle θ.

Joint bond graph: rotational joint between body A and body B, showing velocity constraint bonds and free rotational DOF

See slides 29–31, pack M03

The key practical point is: in the inertial-frame formulation, joint directions $\hat{\mathbf{d}}(\theta)$ are time-varying, making the MTF moduli time-varying. This is why the **body-frame formulation** is typically preferred for multi-body dynamics — joint directions remain constant in the body frame.

6.4 The K.A.M. (Kinematic Analysis Method)

Before addressing closed loops and derivative causality, it is useful to formalize how to build a mechanism's bond graph starting purely from kinematics.

1. **Identify key variables**: Determine the vector of generalized coordinates $\mathbf{q}_k$, the C-field displacements $\mathbf{q}_C$, and the I-field velocities $\mathbf{v}_I$. Note that this method works with velocities (flows).
2. **C-field constraints**: Relate $\mathbf{q}_C$ to $\mathbf{q}_k$ and differentiate to get $\dot{\mathbf{q}}_C = \mathbf{T}_{Ck}(\mathbf{q}_k)\dot{\mathbf{q}}_k$. Write this into the bond graph using MTFs and 0-junctions.
3. **I-field constraints**: Obtain the velocity transformation $\mathbf{v}_I = \mathbf{T}_{Ik}(\mathbf{q}_k)\dot{\mathbf{q}}_k$ and append to the graph via MTFs.
4. **Attach elements**: Append C- and I-fields to their respective junctions. Add dissipation, gravity (as SE), and simplify the MTFs where possible.

This kinematic description uses a junction structure that links flows (assumed to depend on C-energy variables) to couple C-fields and I-fields. However, this procedure often leads to bond graphs with algebraic loops or derivative causality when bodies are rigidly constrained.

6.5 Multi-body problem and constraints

When two rigid bodies are rigidly connected (e.g. a robotic link assembly), the degrees of freedom are reduced by the constraints. A rigid body requires six I-elements, but if rigidly connected, their velocities cannot all be independent. The naive bond graph has *velocity constraints* that force some storage elements into derivative causality, signalling an inconsistency in the model. This is the classic **Multi-Body Problem**.

6.5.1 The "Gaps Method" and Lagrange Multipliers (DAE formulation)

If we treat the constraints mathematically, we define "gap variables" $\boldsymbol{\delta}$ (e.g., horizontal and vertical distances between two theoretically pinned joints) which must vanish ($\boldsymbol{\delta} = \mathbf{0}$). The forces required to enforce these gaps are **Lagrange Multipliers** ($\boldsymbol{\lambda}$). The system is written in a *descriptor form* (a Differential-Algebraic Equation, DAE system), where $E\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$ and matrix E is singular. The Jacobian of the constraints $\boldsymbol{\Delta} = \partial\boldsymbol{\delta}/\partial\mathbf{q}_k$ relates the velocities ($\dot{\boldsymbol{\delta}} = \boldsymbol{\Delta}\dot{\mathbf{q}}_k$) and the constraint forces ($\mathbf{F}_e = \boldsymbol{\Delta}^T\boldsymbol{\lambda}$). In the bond graph, these appear as external forces opposing the natural motion to keep $\dot{\boldsymbol{\delta}} = 0$.

6.5.2 Handling Derivative Causality Physically and Algebraically

To resolve the derivative causality natively in the simulation, two distinct tricks are widely used:

Method 1: The "Parasitic Spring" (Breaking the Constraint). This is a highly practical engineering approach. Physically, two perfectly rigid bodies connected by a rigid joint represent an over-constrained idealisation. In reality, nothing is perfectly rigid; everything flexes. The resolution is to introduce a small **compliance** at the connection. Rather than a rigid bond forcing equal velocities via a strict 1-junction, the joint is modelled as a stiff spring (C-element with large k) and a small damper (R-element) between them. This:

1. Breaks the strict kinematic constraint, restoring all C- and I-elements to integral causality because the masses can now vibrate at slightly different speeds.
2. Introduces a fast time scale (stiff ODE) associated with the high-frequency compliance vibration. Stiff ODE solvers are required.
3. Is physically correct — real connections always have some finite stiffness.

Method 2: The "Matrix Smash" (Using Explicit I-Fields). Instead of letting the computer choke on derivative causality, we use algebra to hide the constraint inside a matrix before simulation. We draw a box around the constrained masses and the MTF hinge connecting them, treating the whole assembly as an *Implicit I-Field*. By computing the combined inertia of the whole arm analytically, we replace the individual masses with one giant, system-wide **Explicit I-Field**. Because this explicit matrix views the robot arm as one unified tensor of momentum rather than competing parts, it operates completely in integral causality. The algebra associated with differential causality is solved once and for all.

6.6 Lagrangian mechanics and IC-fields (D.A.M.)

The combination of I-fields (kinetic energy) and C-fields (potential energy) in a bond graph with a junction structure encoding kinematic constraints is exactly equivalent to a Lagrangian description.

If we have a holonomic, lagrange-regular mechanical system, the kinetic energy is a quadratic-affine form: $T = \frac{1}{2}\dot{\mathbf{q}}_k^T \mathbf{M}(\mathbf{q}_k)\dot{\mathbf{q}}_k + \dot{\mathbf{q}}_k^T \mathbf{a}(\mathbf{q}_k)$. If $\mathbf{q}_k$ are generalised coordinates, then the Lagrange equations:

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i} = E_i \tag{6.13}$$

(where E_i includes potential energy derivatives and external forces) are automatically encoded by the bond graph. The IC-field representation makes this connection explicit and is particularly useful for systems with many degrees of freedom where writing Lagrange equations by hand is error-prone.

The Hamiltonian Form and the D.A.M. The **Dynamic Analysis Method (D.A.M.)** replaces the kinematic junction structure with a true dynamic IC-field based on Hamilton's equations. The Hamiltonian is defined as $\mathcal{H} = \mathbf{p}_k^T \dot{\mathbf{q}}_k - \mathcal{L}$. By defining generalised momenta $\mathbf{p}_k = \partial T / \partial \dot{\mathbf{q}}_k$, we can invert the mass matrix to find velocities:

$$\dot{\mathbf{q}}_k = \mathbf{M}^{-1}(\mathbf{q}_k, t)[\mathbf{p}_k - \mathbf{a}(\mathbf{q}_k, t)] \tag{6.14}$$

And the momentum update is:

$$\dot{\mathbf{p}}_k = \mathbf{e}'_k + \mathbf{E}_k \qquad \text{where } \mathbf{e}'_k = \frac{\partial T}{\partial \mathbf{q}_k} \tag{6.15}$$

In the bond graph, the IC-field acts with an I-part computing $\dot{\mathbf{q}}_k$ (flows) from $\mathbf{p}_k$ (momenta), and a C-part computing the internal efforts $\mathbf{e}'_k$ from $\mathbf{q}_k$ (displacements).

The *main advantage* of this Hamiltonian IC-field approach is that it mathematically guarantees integral causality everywhere, avoiding algebraic loops entirely. The *major drawback* is that the configuration-dependent mass matrix $\mathbf{M}(\mathbf{q}_k, t)$ must be non-singular and continuously inverted during the simulation.

Chapter 7

Models of Distributed Parameter Systems

All models considered so far are **lumped-parameter models**: the state is a finite-dimensional vector, components are point-like, and the number of degrees of freedom is finite. Real structures, however, have **continuously distributed** inertia, compliance, and damping — a beam, a gas column, a hydraulic line. These are described by **partial differential equations** (PDEs), and the number of degrees of freedom is infinite. This chapter develops two strategies for obtaining tractable finite-dimensional bond graph models: *simple lumping* and *separation of variables (normal modes)*.

7.1 Distributed vs. lumped parameter systems

A **distributed-parameter system** is one in which the state (displacement, pressure, temperature, etc.) varies continuously with both time and space. Its model is a PDE of the form:

$$\mathcal{L}\left[\xi(x,t)\right] = f(x,t), \tag{7.1}$$

where $\mathcal{L}$ is a spatial differential operator and f is a forcing term. The PDE is "local" because it describes the system's behavior point by point. A complete problem statement requires:

- The PDE itself (local dynamics).
- **Initial conditions** at $t = 0$ (as many as the order in time).
- **Boundary conditions** at the spatial extremes $x = 0$ and $x = L$ (as many as the order in space).

Only if joined with the proper set of initial and boundary conditions can a PDE be properly solved (yielding a well-posed Cauchy Problem). Notice that a system is typically described with a set of PDEs whose total order is double the space region dimension.

For linear elastic bodies (isotropic materials), the PDE typically takes the **wave equation** form:

$$\frac{\partial^2 \xi}{\partial t^2} = c^2 \frac{\partial^2 \xi}{\partial x^2}, \quad c = \sqrt{E/\rho}, \tag{7.2}$$

with wave speed c determined by the elastic modulus E and density ρ. Deformations propagate through the body as waves — a perturbation at $x = 0$ does not instantaneously influence $x = L$, but arrives after a delay L/c. In a sense, a wave can be mathematically considered as a pair of opposite delays!

Key Insight: When lumped models fail

Lumped models assign all inertia to isolated point masses and all compliance to isolated springs with zero mass. This works well when the excitation frequency ω satisfies $\omega \ll \pi c/L$ (the first natural frequency of the distributed system). When this condition fails — long flexible arms of industrial robots, building floors, transmission lines — distributed effects are essential and lumped models give qualitatively wrong predictions above a certain frequency.

7.2 Simple lumping: discretising the continuum

The most direct approach: **retain finite-size lumps** instead of taking the limit to infinitesimal elements. For a bar of length L, area A, density ρ, and modulus E, divide it into n elements of length $\Delta x = L/n$. Each lump has:

- **Mass**: $m_i = \rho A\, \Delta x$ (I-element).

- **Compliance**: $C_i = \Delta x/(EA)$ (C-element, from Hooke's law $\sigma = E\,\varepsilon$, $\varepsilon = \partial\xi/\partial x \approx (q_i)/\Delta x$).

The generalised momentum and displacement for the i-th lump are:

$$p_i = \rho A\,\Delta x \cdot \dot{\xi}_i \quad \text{(momentum)}, \qquad q_i = \xi_i - \xi_{i-1} \quad \text{(relative deformation)}. \tag{7.3}$$

The equations for each interior lump read:

$$\dot{p}_i = \frac{EA}{\Delta x}(q_{i+1} - q_i), \qquad \dot{q}_i = \frac{1}{\rho A\,\Delta x}(p_i - p_{i-1}). \tag{7.4}$$

These are the **discrete Newton and Hooke laws**: Newton's law is a 1-junction constraint (momentum change = net force); Hooke's law is a C-element (displacement = compliance $\times$ force). The first and n-th elements must be treated carefully since they are boundary elements, typically requiring the addition of S_E or S_F sources to avoid derivative causality at the boundaries.

Key Insight: Electrical equivalent of the bar

There is a direct parallelism with electrical transmission lines. An electrical line can be lumped into elements of length Δx having an inductance $L\Delta x$ and capacitance $C\Delta x$. The equations $\dot{p}_i = \frac{1}{C\Delta x}(q_{i+1} - q_i)$ and $\dot{q}_i = \frac{1}{L\Delta x}(p_i - p_{i-1})$ lead to exactly the same wave equation $\frac{\partial^2 p}{\partial t^2} = \frac{L}{C}\frac{\partial^2 p}{\partial x^2}$ when taking the limit $\Delta x \to 0$.

■ *Bond graph for n-lump bar: chain of alternating I and C elements with 1-junctions, boundary conditions as sources*

See slides 5–8, pack M07

Timoshenko beam. For a beam, two degrees of freedom per cross-section are needed: transverse displacement $w(x,t)$ and rotation $\phi(x,t)$. ϕ is the rotation of the plane cross-section, while θ is the slope of the neutral axis ($\theta \approx \Delta w/\Delta x$). The shear angle is $\gamma = \theta - \phi$.

The Timoshenko model (which accounts for shear deformation and rotary inertia) introduces per lump:

- Translational mass $\rho A\,\Delta x$ and rotary inertia $\rho I\,\Delta x$ (I-elements).
- Bending spring $k_b = EI/\Delta x$ relating moment M to angle difference $\Delta\phi$ (C-element).
- Shear spring $k_s = \kappa GA\,\Delta x^{-1}$ relating shear force V to angle–slope difference $\phi - \Delta w/\Delta x$ (C-element). Here κ takes into account that shear distribution is not perfectly uniform across the section.

Taking the limit $\Delta x \to 0$ recovers the Timoshenko PDEs:

$$\kappa GA\frac{\partial}{\partial x}\left(\frac{\partial w}{\partial x} - \phi\right) = \rho A\frac{\partial^2 w}{\partial t^2}, \tag{7.5}$$

$$EI\frac{\partial^2 \phi}{\partial x^2} + \kappa GA\left(\frac{\partial w}{\partial x} - \phi\right) = \rho I\frac{\partial^2 \phi}{\partial t^2}. \tag{7.6}$$

Neglecting rotary inertia ($\rho I \approx 0$) and assuming $\kappa GA \to \infty$ (infinitely stiff in shear, so $\gamma \to 0$ and $\theta = \phi = \partial w/\partial x$) recovers the simpler **Euler–Bernoulli model**:

$$EI\frac{\partial^4 w}{\partial x^4} + \rho A\frac{\partial^2 w}{\partial t^2} = q(x,t), \tag{7.7}$$

where $q(x,t)$ is the distributed transverse load.



Limitations of simple lumping. The key drawback is the **accuracy–complexity trade-off**: each additional lump improves the prediction of low-frequency behaviour but simultaneously introduces new, highly inaccurate high-frequency modes. A model with n lumps has $2n$ modes, but only the lowest $\sim n/3$ are reasonably accurate. Worse, the stiff high-frequency modes impose severe constraints on integration time step. Also, it is difficult to analytically predict the convergence to the continuous model. This motivates the separation of variables approach.

7.3 Separation of variables and normal modes

Watch out: Major Limitation

The Separation of Variables method works beautifully and provides spectrally exact representations, but it works **ONLY for linear PDEs**. If your system includes nonlinear distributed effects, you are forced to solve the continuum equations numerically or back up into a finite lumping representation.

The idea is to seek solutions of the PDE as products of a spatial function and a temporal function: $\xi(x,t) = Y(x)\,\eta(t)$. Substituting into the unforced wave equation (for the bar):

$$\rho\, Y(x)\,\ddot{\eta}(t) = EA\, Y''(x)\,\eta(t) \quad \Longrightarrow \quad \frac{\ddot{\eta}}{\eta} = \frac{E}{\rho}\frac{Y''}{Y} = -\omega^2, \tag{7.8}$$

where $-\omega^2$ is a separation constant (negative for oscillatory solutions). This yields two **ordinary differential equations**:

$$\ddot{\eta} + \omega^2\eta = 0, \qquad Y'' + k^2 Y = 0, \quad k^2 = \frac{\rho\omega^2}{E}. \tag{7.9}$$

The spatial ODE has general solution $Y(x) = A\cos kx + B\sin kx$; the constants A, B, and the eigenvalues k (and hence natural frequencies ω_n) are determined by the **boundary conditions**.

Example: bar fixed at $x = 0$, free at $x = L$. Boundary conditions: $Y(0) = 0$ (fixed, zero displacement) and $Y'(L) = 0$ (free, zero stress). These give $A = 0$ and $\cos k_n L = 0$, so:

$$k_n L = \left(n - \tfrac{1}{2}\right)\pi, \quad n = 1, 2, 3, \ldots \qquad \omega_n = k_n\sqrt{E/\rho} = \frac{(2n-1)\pi}{2L}\sqrt{\frac{E}{\rho}}. \tag{7.10}$$

The corresponding **mode shapes** (eigenfunctions) are:

$$Y_n(x) = \sin\left(\frac{(2n-1)\pi x}{2L}\right). \tag{7.11}$$

These are the natural shapes in which the bar can vibrate. If the bar is displaced into the shape of the n-th mode and released, it oscillates purely at frequency ω_n.

Orthogonality. Mode shapes of a self-adjoint differential operator are orthogonal with respect to the mass distribution:

$$\int_0^L \rho A\, Y_m(x)\, Y_n(x)\, dx = 0 \quad \text{for } m \neq n. \tag{7.12}$$

This is the key property that decouples the equations.

7.4 Modal decomposition: the forced case and P.F.R.

To handle external forcing within this continuum formulation, we use the **Point-Force Representation (P.F.R.)**. An external active force $F(t)$ acting strictly at a point $\tilde{x}$ is described as a force per unit length using the Dirac delta function: $f(x,t) = F(t)\delta(x-\tilde{x})$.

The response to this force is written as a superposition of all modes:

$$\xi(x,t) = \sum_{n=1}^{\infty} Y_n(x)\,\eta_n(t). \tag{7.13}$$

Substituting this into the forced PDE, multiplying by $Y_m(x)$, and integrating over space $x \in [0, L]$, orthogonality eliminates all cross-terms and yields one decoupled ODE per mode. Because of the Dirac delta integral property ($\int F(t)\delta(x-\tilde{x})Y_m(x)dx = F(t)Y_m(\tilde{x})$), the result is:

$$\boxed{m_n\,\ddot{\eta}_n + k_n\,\eta_n = F(t)\,Y_n(\tilde{x})}, \qquad m_n = \int_0^L \rho A\,Y_n^2\,dx = \frac{\rho A L}{2}, \quad k_n = m_n\omega_n^2. \tag{7.14}$$

This is the **modal equation**: a simple harmonic oscillator with mass m_n (modal mass), stiffness $k_n = m_n\omega_n^2$ (modal stiffness), and forcing $F(t)\,Y_n(\tilde{x})$ (the modal forcing factor evaluates the mode shape exactly at the point where the force is applied). The response is found separately for each mode and then combined.

7.5 Bond graph for the finite-mode model

Introducing modal momentum $p_n = m_n\dot{\eta}_n$ and displacement $q_n = \eta_n$, the modal equation becomes:

$$\dot{p}_n = -k_n q_n + F(t)\,Y_n(\tilde{x}), \qquad \dot{q}_n = p_n/m_n. \tag{7.15}$$

This is exactly the state equation of an I-element ($I = m_n$) and a C-element ($C = 1/k_n$) on a 1-junction, with the external force $F(t)$ connected via a TF of modulus $Y_n(\tilde{x})$. The **finite-mode bond graph** is:

■ *Bond graph for N-mode bar: N parallel I–C pairs (modal oscillators) each connected via TF:Yn(x) to a common port (0-junction) where F is a*

See slides 23–25, pack M07

This is the preferred model because:

1. It captures the exact frequency content up to the retained modes.
2. It has all storage elements in integral causality (when the input is an effort).
3. It can be seamlessly connected to other bond graph subsystems.
4. Retaining N modes gives a model of order $2N$ that is spectrally exact up to ω_N.

Key Insight: Force-free boundary conditions are preferred

When the boundary conditions are uncertain or may change (e.g. a beam whose end attachment varies during operation), it is best to use **force-free mode shapes** — eigenfunctions of the PDE with zero force (and moment) at all boundaries. These modes can represent any motion, including motions at previously "fixed" boundaries (by attaching a stiff spring to ground later), whereas modes derived for a fixed boundary can never generate motion there. Force-free modes are harder to derive but provide a universal modelling library.

Rigid Body Modes. When using force-free boundaries, the frequency equation permits $\omega = 0$. These correspond to **rigid body modes** where the structure moves without deformation. For a free-free beam, there are two such modes: a pure translation ($Y_{00}(x) = 1$) and a pure rotation about the centre of mass ($Y_0(x) = x - L/2$). In the bond graph, these appear as pure I-elements (modal mass and rotary inertia) without accompanying C-elements, because they have zero modal stiffness ($k = 0$).

Reading outputs and applying moments at arbitrary locations. The velocity at any point x_0 along the bar is:

$$\frac{\partial \xi}{\partial t}(x_0, t) = \sum_n Y_n(x_0)\, \dot{\eta}_n(t) = \sum_n Y_n(x_0)\, \frac{p_n}{m_n}. \tag{7.16}$$

In the bond graph, this is a 0-junction connected to TF elements of moduli $Y_n(x_0)$, with each mode's velocity read from its I-element. Similarly, forces can be applied at arbitrary interior points by adding zero-force Se sources at those locations — these serve as output measurement points without injecting any power.

If an external *moment* is applied instead of a force, the associated TF elements coming from the moment 0-junction have the *mode shape slopes* $dY_n(x)/dx$ evaluated at the location of the moment as their moduli, instead of the mode shapes $Y_n(x)$ themselves.

7.6 Damping, Mode Selection, and Causality

Including Damping. The separation of variables formulation mathematically works only without damping. However, physical structures are lightly damped (modal damping ratio $\zeta_i \in [0.01, 0.1]$). Instead of modeling the complex physical damping distribution, we functionally insert damping into the individual modes. The modal equation becomes:

$$m_i \ddot{\eta}_i + b_i \dot{\eta}_i + k_i \eta_i = F_{modal} \tag{7.17}$$

In the bond graph, this is simply an R-element attached to the i-th modal 1-junction, with damping constant $b_i = 2\zeta_i \omega_i m_i$.

How many modes to retain? The Residual Compliance. The number of modes retained depends on the expected frequency content of the inputs. Modes whose natural frequency ω_i is significantly higher than the highest input frequency ω_{max} operate in the **stiffness-controlled region**. In this region, the modal inertia and resistance have virtually no effect. As a rule of thumb, modes are retained up to a frequency 2 to 5 times higher than ω_{max}. For all higher modes, instead of completely pruning them, we can drastically improve the model's static accuracy by consolidating their modal stiffnesses into a single **Residual Compliance**. In the bond graph, all higher-order modal 1-junctions are removed, and a single $C_{residual}$ is placed at the input 0-junction, where $1/k_r = \sum_{i>N}(Y_i^2)/k_i$.

Causality Issues: Flow Inputs. A major problem arises if the external system imposes a velocity (causal flow input) instead of a force. A flow input forces all the TF elements, which in turn forces exactly one I-element of the modes into **derivative causality**. This completely breaks the decoupling of the modal equations and creates a severe algebraic problem involving the derivatives of the inputs.

- **The Fix:** Derivative causality can be structurally avoided by appending an additional modal compliance (without its associated modal inertia) at the flow input. This additional "parasitic" mode absorbs the flow boundary condition and restores integral causality everywhere. While this introduces an algebraic loop among the modes, it is a constant algebraic loop that can be easily solved once and for all inside the model matrix, unlike the massive difficulties of dealing natively with DAE derivative causality.

Assembling complex structures. If a complex structure is composed of many elements (e.g., an A-frame built of three uniform beams), a complete model is constructed by building finite-mode bond graphs for each beam and connecting them at their geometric interfaces using junction structures and TFs. Care must be taken at the connection points, as rigid connections force shared velocities which can again introduce derivative causality, solvable via the compliance techniques described above.

Chapter 8

Thermo-fluid Systems

Thermal and fluid systems require special treatment in the bond graph framework. The fundamental difficulty is that heat (energy exchange due to temperature gradients) does not fit naturally into the standard power-product $e \cdot f$ = power unless the correct thermodynamic variables are chosen. There are two approaches: the simple **pseudo-bond graph** (useful for heat conduction problems but limited) and the rigorous **true bond graph** based on entropy flow as the flow variable.

8.1 Thermodynamic background

Classical vs. irreversible thermodynamics. Classical thermodynamics studies equilibrium states and the energy accounting between them, without modelling the transition process. **Irreversible thermodynamics** models the dynamic evolution — how a system transitions from one state to another via heat flows and work — under the assumption of *local pseudo-equilibrium*: at any instant and for any small volume element, the system is in thermodynamic equilibrium, allowing the use of equilibrium state functions locally.

Internal energy and the Gibbs equation. The **internal energy** per unit mass u of a pure substance depends on entropy per unit mass s and specific volume $v = 1/\rho$:

$$u = u(s, v). \tag{8.1}$$

The fundamental thermodynamic relation (**Gibbs equation**) gives:

$$du = T\,ds - p\,dv, \tag{8.2}$$

where $T > 0$ is temperature (absolute) and $p > 0$ is thermodynamic pressure. In rate form:

$$\dot{u} = T\dot{s} - p\dot{v}. \tag{8.3}$$

The equations of state are the partial derivatives:

$$T = \left.\frac{\partial u}{\partial s}\right|_v, \qquad p = -\left.\frac{\partial u}{\partial v}\right|_s. \tag{8.4}$$

Entropy and disorder. Entropy S quantifies microscopic disorder: the number of possible different microscopic arrangements the particles of a system can assume WITHOUT altering the macroscopic observable state (defined by pressure, temperature, volume). It is macroscopically quantified as $S = k_B \ln \Omega$, where k_B is the Boltzmann constant and Ω is the number of microstates. Heat flow between systems at different temperatures is always spontaneous from high-T to low-T, always irreversible, and always increases total entropy (second law). This irreversibility is the thermodynamic arrow of time. In fact, all physical laws potentially admit time to flow both forward and backward, but time actually always evolves in a forward direction because that is the direction mathematically related to an increase in global entropy.

The thermodynamic laws summary. Before proceeding, it is useful to recall the governing principles:

- **Zeroth Law:** Transitivity of thermal equilibria (if A is in eq. with B, and B with C, then A is in eq. with C). This defines temperature as a universal comparable quantity.
- **First Law:** Energy conservation, embodied in the Gibbs equation $\Delta U = Q - W$.

- **Second Law:** For any possible heat flow, the net and global entropy change is ALWAYS increasing. This defines the irreversibility of time.
- **Third Law:** As $T \to 0$ K, the system's entropy tends to stabilize to a minimum $S^{\min} \geq 0$.

Key Insight: Heat vs. work — the key distinction

Heat $\dot{Q}$ is energy exchange driven by temperature gradients; it changes internal energy via the entropy term $T\dot{S}$. **Work** $\dot{W}$ is energy exchange via macroscopic forces; it changes internal energy via $-p\dot{V}$. The critical difference: work is "ordered" energy (zero entropy increase in an ideal process); heat is "disordered" energy (always increases entropy). In nature, *pure work exchange is never possible* — any macroscopic process generates some entropy increase (dissipation), which appears as heat.

8.2 Pseudo-bond graph for heat transfer

The simplest approach: treat temperature as effort and heat flow rate $\dot{Q}$ as flow. Note that heat flow itself already has the dimensions of power.

$$e \leftrightarrow T, \quad f \leftrightarrow \dot{Q}, \quad \text{but} \quad e \cdot f = T\dot{Q} \neq \text{power}. \tag{8.5}$$

The product $T\dot{Q}$ has units of [K·W] — it is *not* power. This is a **pseudo-bond graph**: it uses bond graph algebra but with non-power variables. Despite this limitation, pseudo-bond graphs give the correct heat transfer equations:

Thermal resistor. Heat flows through a slab of material with thermal conductivity k, thickness l, area A:

$$\dot{Q} = \frac{T_1 - T_2}{R}, \quad R = \frac{l}{kA}. \tag{8.6}$$

In the pseudo-bond graph, this is an R-element with $R = l/(kA)$, connected to two 0-junctions representing the two surface temperatures. The 1-junction implements $T_1 - T_2$.

The missing element: Thermal Inertia. It is crucial to note that **there are no thermal inertias**. Let us suppose a thermal I-element existed. In the pseudo-bond graph, its constitutive law would be $\Delta T = I\frac{d\dot{Q}}{dt}$. This implies that a heat flow $\dot{Q}$ could have a constant, non-zero value even if the temperature difference ΔT is zero, or it could flow against a temperature gradient due to a sort of "thermal momentum". This directly violates the Second Law of Thermodynamics, which postulates that spontaneous heat flow must always be directed from a higher to a lower temperature. Therefore, a thermal I-element cannot exist, as it would imply the possibility of violating fundamental thermodynamics.

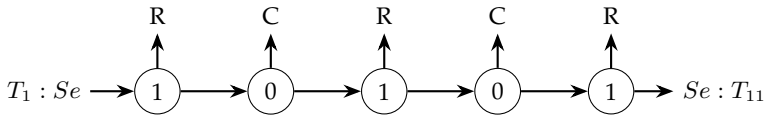
Thermal capacitance. A lump of material with uniform temperature T, specific heat c, and mass m absorbs heat:

$$C\,\dot{T} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}}, \qquad C = mc. \tag{8.7}$$

This is a C-element in the pseudo-bond graph. Causality: $\dot{Q}$ is the flow (input), T is the effort (output) — standard integral causality. The expression for C can be formally derived by assuming the element does negligible work by expanding or contracting (constant volume). From the Gibbs equation $du = T\,ds - p\,dv$, if $dv = 0$, we have $du = T\,ds = dQ/m$. Since the specific heat at constant volume defines $du = c_v dT$, we obtain $dQ = mc_v dT$, or $\dot{Q}dt = mc_v dT$. Integrating this yields $T = T_0 + \frac{1}{mc_v}\int \dot{Q}dt$. Thus, the thermal capacitance is exactly $C = mc_v$.

Watch out: Pseudo-bond graphs cannot couple to true bond graphs

The pseudo-bond graph cannot be directly coupled to a true mechanical or electrical bond graph via power bonds, because $T\dot{Q} \neq$ power. Coupling requires ad-hoc elements (e.g. a mechanical-thermal R-field) that handle the power conversion explicitly. This limitation motivates the true bond graph approach.



Pseudo-bond graph for a pipe wall: alternating R (conduction) and C (thermal mass) elements, analogous to an RC electrical circuit.

8.3 True bond graph for thermodynamics

The correct power variable pair for thermal systems is:

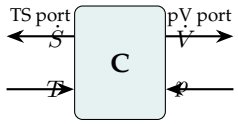
$$e \leftrightarrow T \quad \text{(temperature, effort)}, \qquad f \leftrightarrow \dot{S} \quad \text{(entropy flow rate, flow)}, \qquad P = T\dot{S} = \dot{Q}. \tag{8.8}$$

With this choice, $e \cdot f = T\dot{S}$ *is* the true power (the heat power $\dot{Q}$), since $\dot{Q} = T\dot{S}$ for a reversible heat exchange. This is the **true bond graph** for thermal systems.

The pure substance as a C-field. The Gibbs equation $\dot{u} = T\dot{s} - p\dot{v}$ reveals that a pure substance, viewed as a 2-port element, is a **C-field**:

- Port 1 (thermal): effort T, flow $\dot{S}$, displacement S.
- Port 2 (mechanical): effort p, flow $-\dot{V}$, displacement $-V$ (or $\dot{V}$ with sign absorbed into orientation).

The constitutive law of the C-field is $T = T(S, V)$, $p = p(S, V)$ — the equations of state of the substance. The energy stored in the C-field is the internal energy $U(S, V) = mu(s, v)$.



2-port C-field: pure substance

8.4 Equations of state for a perfect gas

For a perfect (ideal) gas, $pv = RT$ (where R is the specific gas constant) and $u = c_v(T - T_0)$. Using the Gibbs equation to find $u(s, v)$:

From $du = T\,ds - p\,dv$, with $p = RT/v$ and $u = c_vT$:

$$ds = \frac{du}{T} + \frac{p}{T}dv = c_v\frac{dT}{T} + R\frac{dv}{v} \quad \Rightarrow \quad s = c_v \ln\frac{T}{T_0} + R\ln\frac{v}{v_0}. \tag{8.9}$$

Inverting to express T in terms of s and v:

$$\boxed{T(s, v) = T_0\,\exp\left(\frac{s}{c_v}\right)\left(\frac{v}{v_0}\right)^{-R/c_v}} \quad \Rightarrow \quad p(s, v) = -\left.\frac{\partial u}{\partial v}\right|_s = \frac{RT}{v} = p_0\,\exp\left(\frac{s}{c_v}\right)\left(\frac{v}{v_0}\right)^{-\frac{c_p}{c_v}}. \tag{8.10}$$

These are the **complete constitutive equations** for the perfect-gas C-field in integral causality (inputs: $\dot{S}$, $\dot{V}$; outputs: T, p). The constitutive law is nonlinear even for a perfect gas.

Key Insight: Maxwell reciprocal relations for the gas C-field

The Maxwell reciprocal condition $\partial T/\partial(-V)|_S = \partial(-p)/\partial S|_V$ is equivalent to the compatibility of the Gibbs equation — it holds by construction when the constitutive law is derived from a state function $u(s, v)$. Always derive equations of state from a potential (internal energy, enthalpy, Helmholtz or Gibbs free energy) to guarantee this consistency. Arbitrary constitutive laws would not, in general, allow the existence of a conservative internal energy function, potentially allowing net energy production in a cycle (a perpetual motion machine of the first kind).

Legendre transforms and mixed causality. When mixed causality is needed on the C-field (e.g. pressure p as input, volume V as output, while entropy S remains input with temperature T as output), the appropriate potential is the **enthalpy** $h(s, p) = u + pv$:

$$\frac{\partial h}{\partial s} = T, \qquad \frac{\partial h}{\partial p} = v. \tag{8.11}$$

The full set of Legendre transforms:

$$h(s,p) = u + pv \quad \Rightarrow \quad \frac{\partial h}{\partial s} = T,\ \frac{\partial h}{\partial p} = v \quad \text{(enthalpy)} \tag{8.12}$$

$$f(T,v) = u - Ts \quad \Rightarrow \quad \frac{\partial f}{\partial T} = -s,\ \frac{\partial f}{\partial v} = -p \quad \text{(Helmholtz)} \tag{8.13}$$

$$g(T,p) = u + pv - Ts \quad \Rightarrow \quad \frac{\partial g}{\partial T} = -s,\ \frac{\partial g}{\partial p} = v \quad \text{(Gibbs free energy)} \tag{8.14}$$

Each corresponds to a different causal form of the C-field.

8.5 True bond graph for heat transfer: the thermal resistor

In the true bond graph, a thermal resistor is a **2-port R-field**. Two reservoirs at temperatures T_1 and T_2 exchange heat $\dot{Q} = H(T_1 - T_2)$ through the resistor. With $\dot{Q} = T_i \dot{S}_i$:

$$\dot{S}_1 = \frac{\dot{Q}}{T_1} = \frac{H(T_1 - T_2)}{T_1}, \qquad \dot{S}_2 = \frac{\dot{Q}}{T_2} = \frac{H(T_1 - T_2)}{T_2}. \tag{8.15}$$

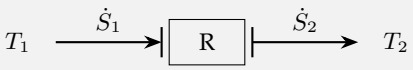
The net entropy production rate is:

$$\dot{S}_2 - \dot{S}_1 = H(T_1 - T_2)\left(\frac{1}{T_2} - \frac{1}{T_1}\right) = H\frac{(T_1 - T_2)^2}{T_1 T_2} \geq 0. \tag{8.16}$$

This is always non-negative (second law of thermodynamics). Unlike the pseudo-bond graph resistor, the true R-field is *not* symmetric in entropy flow: more entropy leaves the hot side ($\dot{S}_1$) than enters the cold side ($\dot{S}_2$) — entropy is generated by the irreversible heat flow. Although the 2-port resistive field is power-conservative ($T_1\dot{S}_1 = T_2\dot{S}_2$), it is **neither a transformer nor a gyrator**, because it strictly increases the entropy of the system independently of the heat flow direction.

Watch out: The only admissible causality for the thermal R-field

The constitutive relation $\dot{S}_i = H(T_1 - T_2)/T_i$ shows that *both temperatures must be inputs* to compute the entropy flows. This means both effort-in causalities are imposed: T_1 and T_2 are inputs, $\dot{S}_1$ and $\dot{S}_2$ are outputs. No other causal form is admissible.

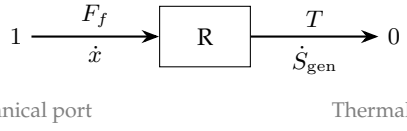


8.6 Thermomechanical coupling

A mechanically dissipative element (e.g. a friction force) dissipates power $\dot{W}_{\text{diss}} = F_f \dot{x} > 0$. In the true bond graph, this power heats the surrounding fluid or structure: $T\dot{S}_{\text{gen}} = F_f \dot{x}$. The dissipated mechanical power enters the thermal C-field via its TS port. This is modelled as a **2-port R-field** connecting the mechanical port $(F, \dot{x})$ to the thermal port $(T, \dot{S})$. The constitutive equations are:

$$F_f = A\,\text{sgn}(\dot{x}) \quad \text{(Coulomb friction)}, \qquad \dot{S}_{\text{gen}} = \frac{|F_f \dot{x}|}{T} = \frac{A|\dot{x}|}{T} \geq 0. \tag{8.17}$$

This models a situation where frictional energy reaches the fluid instantaneously (a standard simplifying assumption).





Interpretation. The bond graph makes visible what energy accounting reveals: when the piston moves, work is done on or by the gas. Some of that work is converted to heat through friction and thermal conduction, increasing entropy. The "lost" mechanical energy is not gone — it has become heat energy stored in the gas and unavailable for mechanical work. The irreversibility is embedded in the R-fields.

8.7 Thermoelectric R-element

A resistor dissipating electrical power at rate ei also converts that power to heat, raising its temperature T. The true bond graph couples the electrical port (e, i) to the thermal port $(T, \dot{S})$:

$$ei = T\dot{S} \quad \Rightarrow \quad \dot{S} = \frac{ei}{T}. \tag{8.18}$$

If $R(T) = R_0(1 + \gamma(T - T_0))$ (temperature-dependent resistance), the constitutive law coupling all four variables is nonlinear. This is the **thermoelectric R-field** — a 2-port element with one electrical port and one thermal port, physically representing the Joule heating effect.

If the resistor's body temperature T is varying and depends on how much thermal energy is stored in it, it must be modeled as a thermal C-element attached to the 0-junction. Assuming no significant power flows at the $p\dot{V}$ port (negligible expansion work against the atmosphere), we use the linear relation $\Delta T = Q/C$. However, we must express T as a function of S. Since $dQ = TdS$, we have $dT = \frac{TdS}{C}$, which integrates to $S = C\ln(T/T_0)$. Inverting this yields the nonlinear constitutive equation for the resistor's thermal capacitance in the true bond graph:

$$T(S) = T_0 e^{S/C}. \tag{8.19}$$

Key Insight: True vs. pseudo in practice

The true bond graph (with $T, \dot{S}$ as thermal port variables) is conceptually correct and allows seamless coupling to mechanical and electrical domains via standard power bonds. The pseudo-bond graph $(T, \dot{Q})$ is simpler and sufficient for pure heat conduction problems with no coupling to other domains. For mixed thermomechanical or thermoelectric systems — which is the case for most real mechatronic components — the true formulation is necessary.

8.8 Fluid dynamic systems revisited: Eulerian approach and Bernoulli

Previously, fluid systems were modeled using a simple 1D lumped parameter approach (Lagrangian perspective). While plausible for short segments and linear elements, estimating the fluid inertia $I = \rho L/A$ has severe limitations for transient conditions, varying density, or varying cross-sectional areas.

A more accurate description requires the **Eulerian approach**, where observations are made at a fixed point in space as different fluid particles pass by. The Eulerian Newton's law relates the velocity field to the pressure gradient and gravity:

$$\rho\frac{Dv}{Dt} = \rho\left(\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial s}\right) = -\frac{\partial p}{\partial s} - \rho g\frac{dz}{ds} \tag{8.20}$$

where D/Dt is the material derivative. For an incompressible fluid, $v(s,t) = Q(t)/A(s)$. Substituting this into the Eulerian Newton's law and integrating along the streamline from s_0 to s_l yields the **Bernoulli equation** for the line element:

$$I\dot{Q} + \frac{\rho Q^2}{2}\left(\frac{1}{A_l^2} - \frac{1}{A_0^2}\right) = p_0(t) - p_l(t) - \rho g(z_l - z_0) \tag{8.21}$$

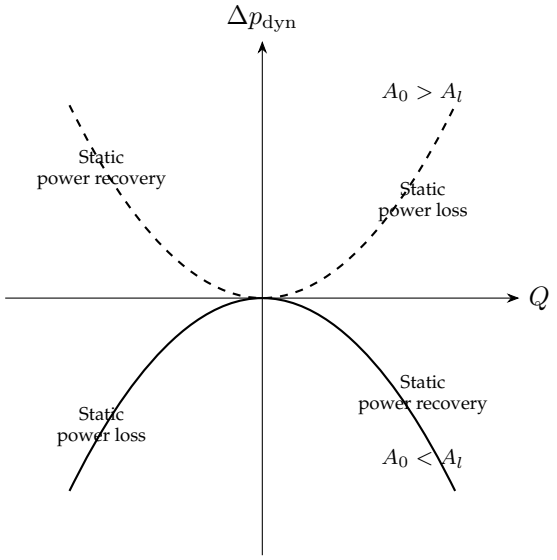
where $I = \rho \int_0^l \frac{ds}{A(s)}$ is the generalized fluid inertia.

The Bernoulli Resistor (R_B). The term involving Q^2 represents the dynamic pressure associated with the kinetic energy of the fluid. In a bond graph, it is represented as an unusual resistance, the **Bernoulli resistance** R_B.

$$\Delta p_{\mathrm{dyn}} = \frac{\rho Q^2}{2}\left(\frac{1}{A_l^2} - \frac{1}{A_0^2}\right) \tag{8.22}$$

Unlike standard resistors, R_B does not always dissipate power. It models the conversion between static pressure energy and kinetic energy (dynamic pressure).

- If $A_0 > A_l$ (a nozzle), the fluid accelerates, kinetic energy increases, and static pressure drops. This operates in the **static power loss** region.
- If $A_0 < A_l$ (a diffuser), the fluid decelerates, kinetic energy decreases, and static pressure is recovered. This operates in the **static power recovery** region, where the resistance appears to *supply* static power.



Constitutive curves of the Bernoulli Resistor showing static power loss and recovery regions.

8.9 Pseudo-bond graphs for compressible gases

When dealing with compressible gases where density changes drastically, maintaining a true bond graph representation becomes computationally complex because the Eulerian description entangles inertial and compliance aspects. This motivates the **thermodynamic accumulator**, a specialized pseudo-bond graph element.

The control volume transports mass, so the flow variables become mass flow rate $\dot{m}$ and enthalpy flow rate $\dot{E}_h$. The accumulator acts as a multi-port C-field with integral causality, storing internal energy E and mass m:

$$\dot{E} = \sum \dot{E}_{h,\mathrm{in}} - \sum \dot{E}_{h,\mathrm{out}} - p\dot{V}, \qquad \dot{m} = \sum \dot{m}_{\mathrm{in}} - \sum \dot{m}_{\mathrm{out}} \tag{8.23}$$

This formulation smoothly interfaces with restrictors (modeling isentropic nozzles) to build complete, low-order, accurate models of interacting thermo-fluid systems, such as piston-ported engines or air cylinders with leakage.